THE SCHUBERT BIALGEBRA AND THE RING OF RC GRAPHS

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1. Introduction

The Schubert bialgebra and its dual. In this article we introduce a bialgebra $\mathcal{A}$ that we call the Schubert bialgebra. This bialgebra has a basis of Schubert polynomials $\mathfrak{S}_u^n$ for $u \in S_\infty$, where the last descent of u is not past index n. The structure constants for multiplication in this basis are the same as the structure constants for multiplication of Schubert polynomials. The difference from the usual realm of study of Schubert polynomials, which is the polynomial ring in infinitely many variables, is that this algebra is graded into length components that interact trivially. That is,

$$\mathcal{A} = \bigoplus_{n=0}^{\infty} \mathbb{Z}[x_1, \dots, x_n]$$

This allows us to consistently define a "splitting" coproduct that is compatible with the usual product of polynomials but actively requires the triviality of products between length components to achieve a bialgebra structure.

We focus primarily on the dual $\mathcal{D}$ of this algebra, where the product is derived from this splitting (deconcatenation) coproduct. Specifically, let Comp_n be the set of weak compositions of length n. Then, additively,

$$\mathcal{D} = \bigoplus_{n=0}^{\infty} \mathbb{Z}^{\oplus \mathrm{Comp}_n}$$

and multiplicatively the product of two basis elements indexed by $a \in \mathrm{Comp}_p$ and $b \in \mathrm{Comp}_q$ is the basis element indexed by the concatenation of a and b, which is an element of Comp_{p+q}. The coproduct on $\mathcal{D}$ is given by splitting sums of exponents, that is to say

$$\Delta(c) = \sum_{a+b=c} a \otimes b$$

where a and b are weak compositions of lengths p and q respectively, and the equation $a + b = c$ indicates that the pointwise sum of a and b is equal to c. The counit is given by $\varepsilon(c) = 0$ unless c is the empty composition, in which case $\varepsilon(c) = 1$. With these definitions, $\mathcal{A}$ and $\mathcal{D}$ are dual bialgebras over $\mathbb{Z}$, a fact that is straightforward to verify. They fail to be connected and are not Hopf algebras.

Combinatorially, $\mathcal{D}$ is interesting, but not fundamentally new at this point. What is new is the Schubert basis Ξ_u^n of $\mathcal{D}$ indexed by $u \in S_\infty$ and n at least as large as the last descent of u, defined by

$$\langle \Xi_u^m, \mathfrak{S}_v^n \rangle = \delta_{u,v}\delta_{m,n}$$

where we take $\langle -, - \rangle$ to be the pairing implied by the assertion that for a composition $c \in \mathrm{Comp}_n$ we have that

$$\langle c, x_a \rangle = \delta_{c,a}$$

where a is any composition. We present an explicit formula for the dual Schubert basis in $\mathcal{D}$ with integer coefficients. These are extremely similar to the dual Schubert polynomials of Postnikov and Stanley defined in [26], and in fact are related by a scalar factor and a trivial change of basis. The dual Schubert polynomials of Stanley and Postinkov however do not naturally posses the multiplicative structure of $\mathcal{D}$, but do come with an integral formula that mimics the action of divided difference operators on Schubert polynomials. In principle these structures should be transferrable to our dual Schubert basis, but we have not yet explored this.

The Littlewood-Richardson rule for dual Schubert polynomials. The reader familiar with the landscape may already see that the Schubert basis of $\mathcal{D}$ has nonnegative structure constants. This was observed by Macdonald [19] as well as Bergeron and Billey [3], with a more general result proved by Bergeron and Sottile in [4]. Our main result is a positive, combinatorial formula for the structure constants for multiplication in the dual Schubert basis, that is to say a Littlewood-Richardson type rule for these structure constants, the numbers $d_{u,v}^{w}(p,q)$ such that

$$\Xi_u^p \Xi_v^q = \sum_w d_{u,v}^w(p,q)\Xi_w^{p+q}$$

The combinatorial objects that we use to express this formula are RC graphs, the definition of which can be found, for example, in [3]. The reader may be more familiar with the term "pipe dreams," introduced as a catchier name for the same objects by Knutson and Miller in [14]. We retain the name RC graphs since this method of approaching the objects (thought of as reduced words paired with compatible sequences, with the generators arranged in rows) is far more tractable for our purposes, most of the time. The original intention was indeed to model such biwords, initially presented as a positive formula for Schubert polynomials in [6].

It is important to note that while it common to consider two RC graphs to be the same if they have the same set of crossings, we add an datum, the number of rows, that can distinguish two RC graphs that would otherwise be considered identical. It is a key feature in this exposition that we limit our focus to cases with finitely many rows/variables.

For an RC graph R with n rows, we define two associated RC graphs $\mathtt{clip}^p(R)$ and $\mathtt{trim}^p(R)$ such that $\mathtt{trim}^p(R)$ is obtained directly from the last $n-p$ rows of R, whereas $\mathtt{clip}^p(R)$ is obtained by an insertion algorithm that combinatorially models the Lascoux-Schützenberger transition formula for Schubert polynomials, the latter of which is expounded upon in detail by Macdonald in [19], with the algorithm maintaining numerous structures that can be imposed on RC graphs. We point out the definition of the function $\mathcal{Z} : \mathcal{RC}_n^0 \to \mathcal{RC}_{n-1}$ in Definition 4.3.1, where $\mathcal{RC}_n^0$ indicates the set of RC graphs with n rows such that row n is empty, that realizes the transition formula, for discussion below. Our main theorem is the following.

Theorem 1.1. *Suppose $u, v \in S_\infty$ and $p, q \geq 0$ are integers such that the last descent of u is at most p and the last descent of v is at most q. Define integers $d_{u,v}^w(p,q)$ by the equation*

$$\Xi_u^p \Xi_v^q = \sum_w d_{u,v}^w(p,q)\Xi_w^{p+q}$$

Then for any fixed choices of RC graphs $U \in \mathcal{RC}_p(u)$, $V \in \mathcal{RC}_q(v)$, and a permutation $w \in S_\infty$, we have that $d_{u,v}^w(p,q)$ is the number of RC graphs $R \in \mathcal{RC}_{p+q}(w)$ such that $\mathtt{clip}^p(R) = U$ and $\mathtt{trim}^p(R) = V$, and is in particular nonnegative.

We note that, when restricted to Grassmannian permutations, Theorem 1.1 reduces directly to a Littlewood–Richardson rule for Schur polynomials: if u, v, w are Grassmannian with unique descents at p, q, $p+q$ corresponding to partitions λ, μ, ν, then Ξ_u^p, Ξ_v^q, Ξ_w^{p+q} are precisely the Schur polynomials $s_\lambda(x_1, \ldots, x_p)$, $s_\mu(x_{p+1}, \ldots, x_{p+q})$, $s_\nu(x_1, \ldots, x_{p+q})$, and the classical splitting

$$s_\nu(x_1, \ldots, x_{p+q}) = \sum_{\lambda,\mu} c_{\lambda,\mu}^\nu \, s_\lambda(x_1, \ldots, x_p) \, s_\mu(x_{p+1}, \ldots, x_{p+q})$$

identifies $c_{\lambda,\mu}^\nu = d_{u,v}^w(p,q)$. Theorem 1.1 thus gives an alternative positive combinatorial rule for the classical Littlewood–Richardson coefficients in terms of RC graphs and the $\mathtt{clip}^p/\mathtt{trim}^p$ decomposition.

Key polynomials. The *key polynomials* (also called *Demazure characters* in type A) form a $\mathbb{Z}$-basis $\{\kappa_a\}$ of the polynomial ring $\mathbb{Z}[x_1, x_2, \ldots]$ indexed by weak compositions a. Their origin lies in the work of Demazure [7, 8], who introduced isobaric divided difference operators π_i in order to compute the characters of the B-modules $H^0(X_w, \mathcal{L}_\lambda)$ associated to Schubert varieties X_w in G/B. In type A, the resulting characters become polynomials in the x_i, and one obtains κ_a for any weak composition a by setting $\kappa_\lambda = x^\lambda$ when λ is a partition (an antidominant weight in the convention used here) and applying the appropriate operator π_i at each descent of a to interpolate between rearrangements of λ.

A purely combinatorial development of key polynomials was carried out by Lascoux and Schützenberger in [16], where they introduced the notion of the *right key* $K_+(T)$ and *left key* $K_-(T)$ of a semistandard

Young tableau T, and proved that

$$\kappa_a = \sum_T x^{\mathtt{wt}(T)}$$

where the sum runs over semistandard Young tableaux T of shape sort(a) whose right key $K_+(T)$ is bounded above (entrywise) by the key tableau associated to a. They additionally proved that Schubert polynomials decompose as a positive sum of key polynomials, which was given a combinatorial proof in terms of compatible sequences and the right-key map by Reiner and Shimozono [27]. The intermediate basis of *Demazure atoms* (the "standard bases" of [16]) was later given an explicit nonnegative monomial expansion by Mason [20] via semi-skyline augmented fillings.

From a representation-theoretic perspective, Kashiwara [13] introduced *Demazure crystals*: for a dominant integral weight λ with crystal $B(\lambda)$ and a Weyl group element w with reduced expression $w = s_{i_1} \cdots s_{i_\ell}$, the Demazure crystal $B_w(\lambda) \subseteq B(\lambda)$ is defined inductively from the highest weight element by closure under the operators $\{f_{i_k}^n : n \geq 0\}$, and Kashiwara proved that its character coincides with the Demazure character. Littelmann [17] obtained parallel results in the path model. In type A, this means κ_a is precisely the character of a Demazure crystal inside a tensor product of fundamental crystals, and the key polynomial expansion of any nonnegative-on-Demazure-crystals polynomial is forced to be nonnegative. Assaf and Schilling [1], building on the crystal on reduced factorizations of Morse and Schilling [22], exhibited an explicit Demazure crystal structure on the set of RC graphs (equivalently, pipe dreams) of a permutation w, thereby giving a representation-theoretic proof of the Lascoux–Schützenberger key-positivity of Schubert polynomials. It is this Demazure crystal structure on RC graphs that we will repeatedly invoke below.

There is a dual key basis $\{\kappa_a^*\}$ of $\mathcal{D}$ defined by $\langle \kappa_a^*, \kappa_b \rangle = \delta_{a,b}$, where as before we are restricting to fixed lengths of compositions, and we give an explicit formula for these dual key polynomials in terms of RC graphs. Specifically, for an RC graph R, we define the *extremal weight* extwt(R) to be the extremal weight of the Demazure crystal of R, which is the unique lowest weight element that generates the crystal, and we define Dem(R) to be the set of RC graphs in the same Demazure crystal as R.

Our $\mathcal{Z}$ has an alternative description in terms of Little bumps on reduced words, as defined by Little in [18]. It was shown by Hamaker and Young [11] that Little bumps preserve Coxeter-Knuth moves, which precisely characterize the Demazure crystals of an RC graph, so that $\mathcal{Z}$ preserves the Demazure crystal structure on RC graphs. This allows us to give a Littlewood-Richardson type rule for multiplication in the dual key basis. Specifically, we have the following.

Theorem 1.2 (LR rule for dual keys)**.** *Let a, b be weak compositions of lengths p and q, respectively. Define integers $k_{a,b}^c$ for weak compositions c of length $p+q$ by the equation*

$$\kappa_a^* \kappa_b^* = \sum_c k_{a,b}^c \kappa_c^*$$

Fix an RC graph C such that extwt$(C) = c$*. Then $k_{a,b}^c$ is the number of distinct $R \in$* Dem(C) *such that* extwt$(\mathtt{clip}^p(R)) = a$ *and* extwt$(\mathtt{trim}^p(R)) = b$*, and is in particular nonnegative.*

Equivalently, Theorem 1.2 is a positive combinatorial branching rule

$$\kappa_c(x_1, \ldots, x_{p+q}) = \sum_{a,b} k_{a,b}^c \, \kappa_a(x_1, \ldots, x_p) \, \kappa_b(x_{p+1}, \ldots, x_{p+q})$$

expressing a key polynomial in $p+q$ variables as a nonnegative integer combination of products of key polynomials in the first p and last q variables. The coefficients $k_{a,b}^c$ may be regarded as the Demazure analogue of the $GL_{p+q} \downarrow GL_p \times GL_q$ Littlewood–Richardson branching coefficients: in the full-Demazure (Schur) case they reduce to the classical $c_{\mu,\nu}^\lambda$. Their nonnegativity is, in retrospect, a consequence of the theory of *excellent filtrations* for B-modules introduced by Polo [25] and van der Kallen [32], with existence of such filtrations on tensor products of Demazure modules established in general by Mathieu [21]; see also Joseph [12]. Restricting a Demazure module $V_c(\lambda)$ for GL_{p+q} to the Borel of the Levi $GL_p \times GL_q$ yields a B-module with an excellent filtration whose Demazure subquotients account for the $k_{a,b}^c$. While this abstract nonnegativity is classical, no explicit positive combinatorial rule for these coefficients appears in the literature to our knowledge, and the dual key basis itself does not appear to have been studied before.

Forest polynomials. Note that *any* polynomial basis that is graded by length and degree has a corresponding dual basis in $\mathcal{D}$. In particular, the *forest polynomials* are such a basis. These were introduced by Nadeau and Tewari in [23] and are closely related to Schubert polynomials. They can be defined by reduced words and compatible sequences, but the invariant preserved by these words is distinct from the permutation. Given a reduced word/compatible sequence pair $(\mathbf{r}, \mathbf{s})$, there is an insertion algorithm that produces a corresponding invariant $\mathcal{F}(\mathbf{r}, \mathbf{s})$ that represents an equivalence relation on the words, where the compatible sequence can vary arbitrarily. We therefore shorten the notation to $\mathcal{F}(\mathbf{r})$. The equivalence classes $\mathcal{F}$ consist of words that are all for the same permutation, so that if we define

$$R \equiv_{\mathfrak{P}} R'$$

to mean that $\mathcal{F}(\mathrm{word}(R)) = \mathcal{F}(\mathrm{word}(R'))$, then the forest polynomials partition the set of RC graphs for the given permutation. The forest polynomials are indexed by weak compositions, so we can define a weak composition $\mathfrak{c}_f(R)$ by associating an indexed forest to this equivalence class as in the article. Define $\mathcal{C}_{\mathfrak{P}}(a)$ for a composition a to be the set of equivalence classes of RC graphs under this relation such that $\mathfrak{c}_f(R) = a$.

Theorem 1.3 (LR rule for dual forests)**.** *Let a, b be weak compositions of lengths p and q, respectively. Define integers $f^c_{a,b}$ for weak compositions c of length $p+q$ by the equation*

$$\mathfrak{P}^*_a \mathfrak{P}^*_b = \sum_c f^c_{a,b} \mathfrak{P}^*_c$$

Fix an RC graph C such that $\mathfrak{c}_{\mathfrak{P}}(C) = c$. Then $f^c_{a,b}$ is the number of distinct $R \in [C]$ such that $[\mathtt{clip}^p(R)] \in \mathcal{C}_{\mathfrak{P}}(a)$ and $[\mathtt{trim}^p(R)] \in \mathcal{C}_{\mathfrak{P}}(b)$, and is in particular nonnegative.

As with the dual key basis, Theorem 1.3 is equivalent to a positive combinatorial branching formula

$$\mathfrak{P}_c(x_1, \ldots, x_{p+q}) = \sum_{a,b} f^c_{a,b}\, \mathfrak{P}_a(x_1, \ldots, x_p)\, \mathfrak{P}_b(x_{p+1}, \ldots, x_{p+q})$$

expressing a forest polynomial in $p+q$ variables as a nonnegative integer combination of products of forest polynomials in the first p and last q variables. To the best of our knowledge, no such positive combinatorial splitting rule for forest polynomials has previously appeared in the literature, and the dual forest basis itself does not appear to have been studied before.

The ring of RC graphs. Perhaps the strongest indication that this is the "correct" way to decompose RC graphs is that these formulas are proved by the fact that if we define

$$\mathcal{BRC} = \bigoplus_{n=0}^{\infty} \mathbb{Z}^{\oplus \mathcal{RC}_n}$$

then there is an associative product $\diamond$ on $\mathcal{BRC}$ and a homomorphism of rings $\alpha : \mathcal{BRC} \to \mathcal{D}$ such that

$$\alpha(R) = \Xi^n_u$$

where u is the permutation of R and n is the number of rows. Indeed, $\mathcal{BRC} \cong \mathcal{D} \ltimes \Omega$ as rings, where Ω is the kernel of α (Theorem 4.6). The product $\diamond$ is defined directly with `clip` and `trim`, the former based directly on the transition function $\mathcal{Z}$, and the fact that α is a homomorphism of rings is essentially equivalent to Theorem 1.1. Theorems 1.2 and 1.3 piggyback off of this as well, defining a ideals Θ and Γ of $\mathcal{BRC}$ such that $\mathcal{BRC}/\Theta$ is indexed by Demazure crystals of RC graphs and $\mathcal{BRC}/\Gamma$ is indexed by forest equivalence classes of RC graphs while maintining the projection to $\mathcal{D}$ and leveraging this to find the formula.

Conjectures. Beyond this, there is potentially much to explore. For example, the coproduct of $\mathcal{D}$ has as its structure constants the structure constants of Schubert polynomials, which have evaded expression in terms of positive formulas for decades, and at the time of this writing no positive, combinatorial formula is known for the structure constants of Schubert polynomials that applies in all cases. At the very least, this coproduct is a new potential method of attack on this and related problems, since as far as we know it is new. An expression for Ξ^n_w in the composition basis leads to a simple, purely mechanical method for computing all $c^w_{u,v}$ (generalized Littlewood-Richardson coefficients for Schubert polynomials) by leveraging the extremely simple form the coproduct takes on the weak composition basis, as well as a neat positive formula for the expansion of a weak composition $[a]$ in the Schubert basis.

For the reader familiar with bumpless pipe dreams [15], we state without proof that our apparently complex transition algorithm for RC graphs can be realized almost trivially for bumpless pipe dreams. Specifically, while (reduced) bumpless pipe dreams are typically defined in such a way that they cannot deviate from a square shape, there is indeed a consistent way to remove rows from a bumpless pipe dream unambiguously. This was observed by Yu in [34, Lemma 3.18] in the definition of a k-row BPD. Wiengandt in [33] defines a transition formula $\mathbf{t} : \mathrm{BPD}_n \to \mathrm{BPD}_{n-1}$ and while the image is realized as square, it agrees with Yu's notion of an $n-1$-row BPD.

Gao and Huang in [9] defined a natural bijection between RC graphs and bumpless pipe dreams. Given a bumpless pipe dream B with n rows, we define $\phi(B)$ to be the corresponding RC graph under this bijection.

Conjecture 1.4. *Let B be a k-row reduced bumpless pipe dream such that row k has no blanks ("weighty tiles"). Then*

$$\mathscr{Z}(\phi(B)) = \phi(\mathbf{t}(B))$$

2. The Schubert Bialgebra and its dual

2.1. Some preliminaries.

Definition 2.1.1 (Two Pieri relations $\xrightarrow{k}$ and $\xRightarrow{k}$)**.** The relations $\xrightarrow{k}$ and $\xRightarrow{k}$ are defined on S_∞ as follows, with different notation, in [31]. For $v, w \in S_\infty$, we have $v \xrightarrow{k} w$ if there is a sequence of transpositions $t_{a_1 b_1}, \ldots, t_{a_m b_m}$ such that the following hold:

(1) $a_i \leq k < b_i$ for all i.
(2) $\ell(ut_{a_1b_1} \cdots t_{a_ib_i}) = \ell(u) + i$ for all i.
(3) $w = ut_{a_1b_1} \cdots t_{a_mb_m}$.
(4) The integers $a_1, \ldots, a_m$ are pairwise distinct.

For $\xRightarrow{k}$, only (4) changes:

(4') The integers $b_1, \ldots, b_m$ are pairwise distinct.

Notice that necessarily $m \leq k$ for $\xrightarrow{k}$, but there is no upper bound on m for $\xRightarrow{k}$. This reflects the fact that the following theorem holds:

Theorem 2.1 ([31, Theorem 1])**.** *Let $p \geq 0$, $k > 0$ be integers and let u be a permutation.*

I. *We have the formula*
$$\mathfrak{S}_u(x) h_{p,k}(x) = \sum_{\substack{u \xRightarrow{k} w \\ \ell(u,w)=p}} \mathfrak{S}_w(x)$$
where $h_{p,k}(x)$ is the complete homogeneous symmetric polynomial of degree p in the first k variables.

II. *We have the formula*
$$\mathfrak{S}_u(x) e_{p,k}(x) = \sum_{\substack{u \xrightarrow{k} w \\ \ell(u,w)=p}} \mathfrak{S}_w(x)$$
where $e_{p,k}(x)$ is the elementary symmetric polynomial of degree p in the first k variables.

Definition 2.1.2 (Pulling out variables from double Schubert polynomials, the relation $\overset{i}{\searrow}$)**.** For a permutation v' such that $v' \in S_n$, define

$$\varphi_{i,n}(v')(j) = \begin{cases} v'(j) & \text{if } j < i \\ n+1 & \text{if } j = i \\ v'(j-1) & \text{if } i < j \leq n+1 \\ j & \text{if } j > n+1 \end{cases}$$

Now fix $v \in S_n$ and $i \geq 1$ an integer, we define a relation $\overset{i}{\searrow}$ by declaring that $v \overset{i}{\searrow} v'$ if

$$v \xRightarrow{i} \varphi_{i,n}(v')$$

Equivalently, $v \overset{i}{\searrow} v'$ if whenever $v \in S_n$ and n is minimal, we have that v satisfies the relation $\overset{i}{\Rightarrow}$ with respect to the permutation obtained from v' by inserting $n+1$ at position i. We note that this concept was introduced by Bergeron and Sottile in [5].

If $v \overset{i}{\searrow} v'$, we define a set of integers $Q_i(v', v)$ by

$$Q_i(v', v) = \{v(j) \mid j > i \text{ and } v'(j-1) = v(j)\}$$

Given the fact that any element of S_∞ fixes all but finitely many positive integers, it follows that $Q_i(v', v)$ is a finite set.

The permutations v' such that $v \overset{i}{\searrow} v'$ arise when pulling a variable out of a Schubert polynomial and expressing the coefficients of the powers of this variable as Schubert polynomials in the remaining variables, as is done in [4], from which this definition essentially comes.

For convenience, if a is a weak composition of length n and $v, v' \in S_\infty$, we declare that $v \overset{a}{\searrow} v'$ if

$$v \overset{a_1}{\searrow} v_1 \overset{a_2}{\searrow} v_2 \overset{a_3}{\searrow} \cdots \overset{a_n}{\searrow} v'$$

Definition 2.1.3 (Upward shifting of permutations, maximum right descent)**.** For a permutation $v \in S_\infty$, we define $\uparrow v$ to be the permutation defined by

$$\uparrow v(i) = \begin{cases} v(i-1)+1 & \text{if } i > 1 \\ 1 & \text{if } i = 1 \end{cases}$$

Recursively, we define $\uparrow^k(v)$ to be $\uparrow(\uparrow^{k-1}(v))$. In the literature, this is sometimes written as

$$\uparrow^k v = 1^k \times v$$

For a permutation w, we define $\mathfrak{m}(w)$ to be the maximum right descent of w. That is

$$\mathfrak{m}(w) = \max\{i \mid w(i) > w(i+1)\}$$

The next proposition specializes, in the case of ordinary Schubert polynomial $\mathfrak{S}_v(x)$, to a formula that isolates a chosen index i: one can express $\mathfrak{S}_v(x)$ as a sum of terms of the form $x_i^p\, \mathfrak{S}_{v'}(x^{(i)})$, thereby effectively extracting the variable x_i and leaving Schubert polynomials in the remaining variables [5, Theorem 5.1]. Proposition 2.1.4 is the double Schubert polynomial version of this, which, as far as we know, is new.

Proposition 2.1.4. *Let $v \in S_\infty$ and let $i > 0$ be an integer. Then we have*

$$\mathfrak{S}_v(x; y) = \sum_{v \overset{i}{\searrow} v'} \mathfrak{S}_{v'}(x^{(i)}; y) \prod_{b \in Q_i(v', v)} (x_i - y_b)$$

See the Appendix for a proof.

2.2. **Definition.** We define a commutative algebra $\mathcal{A}$ over the integers as follows. For each n, define $\mathcal{A}_n$ to be the polynomial ring over $\mathbb{Z}$ in the variables $x_1, \ldots, x_n$. Then define

$$\mathcal{A} = \bigoplus_{n=0}^{\infty} \mathcal{A}_n$$

The multiplication within $\mathcal{A}_n$ is as usually defined for the polynomial ring. However, if $a \in \mathcal{A}_m$ and $b \in \mathcal{A}_n$ with $m \neq n$ and $m, n > 0$, then

$$ab = 0$$

Otherwise, the component for $n = 0$ is identified with the coefficient ring. Note that the "identity element" for positive n is not an identity element of $\mathcal{A}$ (we may think of it as a "fat identity").

Each $\mathcal{A}_n$ has a basis consisting of elements x_a, where a is a weak composition of length n, and the notation indicates that

$$x_a = x_1^{a_1} \cdots x_n^{a_n}$$

The direct sum therefore has a basis that can canonically be identified with union of these.

We define a coproduct $\Delta : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ on the basis $x_c^{(n)}$ by

$$\Delta(x_c) = \sum_{ab=c} x_a \otimes x_b$$

where the equation $ab = c$ indicates that a concatenated with b is equal to c. We also define a counit $\varepsilon : \mathcal{A} \to \mathbb{Z}$ by $\varepsilon(x_a) = 0$ unless a is the empty composition.

Lemma 2.2.1. *With Δ and ε, $\mathcal{A}$ is a coassociative, counital coalgebra.*

Proof. We have

$$\Delta(x_d^{(n)}) = \sum x_a^{(p)} \otimes x_c^{(q)}$$

Applying Δ to either tensor factor results in

$$\sum x_a^{(p)} \otimes x_b^{(q)} \otimes x_c^{(r)}$$

The symmetry of this is exactly the coassociativity condition. Seeing that we may choose $p = n$ or $q = n$, the definition of the counit gives us the result that $\mathcal{A}$ is counital as well under Δ and ε. □

Lemma 2.2.2. $\Delta : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ *is a homomorphism of rings.*

Proof. This is where the condition that $x_a^{(p)} x_b^{(q)} = 0$ unless $p = q$ when both $p, q > 0$ comes in. It ensures that only monomials of the same length have nonzero products and preserves the structure of the coproduct as a homomorphism of rings. □

Lemma 2.2.3. $\nabla : \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ *is a homomorphism of coalgebras.*

Proof. Consider the basis element $x_a \otimes x_b$ of $\mathcal{A} \otimes \mathcal{A}$. The product is

$$x_{a+b}$$

Applying Δ results in

$$\sum_{a'b'=a+b} x_{a'} \otimes x_{b'}$$

Conversely, applying $\Delta \otimes \Delta$ yields

$$\sum_{a_1a_2=a, b_1b_2=b} x_{a_1} \otimes x_{a_2} \otimes x_{b_1} \otimes x_{b_2}$$

and applying $\nabla \otimes \nabla$ yields

$$\sum x_{a_1+b_1} \otimes x_{a_2+b_2}$$

The definition of ∇ ensures that a_1 and b_1 are the same length for a nonzero term, and similarly for a_2 and b_2. Setting $a_1 + b_1 = a'$, $a_2 + b_2 = b'$ yields the result. □

Corollary 2.2.4. *$\mathcal{A}$ is a bialgebra over $\mathbb{Z}$.*

$\mathcal{A}$ is afforded a grading into finite dimensional components by observing that each $\mathcal{A}_n$ itself is a graded ring with each homogeneous component being a finitely generated free module. Considering the pair (n, d), where n is the number of variables and d is the degree, as a $\mathbb{Z}^2$ grading, we may take the graded dual module $\mathcal{D}$, which, by virtue of the grading, is isomorphic as a free module to $\mathcal{A}$. We identify the dual basis element x_a^* with the sequence of nonnegative integers a. That is to say,

$$\langle a, x_b \rangle = \delta_{ab}$$

Inspection of the coproduct reveals that the product of a and b in $\mathcal{D}$ is simply the concatenation ab. Hence $\mathcal{D}$ is isomorphic to the free associative algebra on a countable set indexed by nonnegative integers.

$\mathcal{D}$ also has a coproduct compatible with its product, namely

$$c \mapsto \sum_{a+b=c} a \otimes b$$

Thus $\mathcal{A}$ and $\mathcal{D}$ are dual bialgebras.

Calling $\mathcal{A}$ the "Schubert bialgebra" may seem unnecessarily grandiose, however the reason will become clear below.

2.3. **The Schubert basis.** In $\mathcal{A}$, each $\mathcal{A}_n$ has a basis of Schubert polynomials $\mathfrak{S}_u^{(n)}$, where the largest right descent of u is at most n, ensuring that $\mathfrak{S}_u(x)$ has at most n variables. Schubert polynomials limited to a specific number of variables have well-defined structure constants $c_{u,v}^w$, independent of n, given by

$$\mathfrak{S}_u^{(n)}\,\mathfrak{S}_v^{(n)} = \sum_w c_{u,v}^w\,\mathfrak{S}_w^{(n)}$$

These are known to be nonnegative integers by virtue of the fact that they count points of transverse intersections of generic translates of Schubert varieties, however except in special cases no positive combinatorial formula is known.

2.4. **The elementary basis.** An n-elementary weak composition is a finite sequence of nonnegative integers $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_N)$ such that $\alpha_i \leq i$ for all $i < n$ and $\alpha_i \leq n$ for all $i \geq n$, satisfying the additional restriction that $\alpha_i \geq \alpha_{i+1}$ if $i \geq n$.

An n-elementary monomial is a an element of $\mathcal{A}_n$ of the following form:

$$e_{\alpha_1}^1 \cdots e_{\alpha_n}^n e_{\alpha_{n+1}}^n \cdots e_{\alpha_m}^n$$

where α is an n-elementary weak composition. Define $\mathcal{E}_n$ to be the set of n-elementary monomials.

Proposition 2.4.1. *$\mathcal{E}_n$ forms a $\mathbb{Z}$-basis for $\mathcal{A}_n$, and $\bigcup_n \mathcal{E}_n$ forms a $\mathbb{Z}$-basis for $\mathcal{A}$.*

Proof. It is a theorem of Macdonald that the polynomial ring $\mathbb{Z}[x_1, \ldots, x_n]$ is a free module over Λ_n with basis the Schubert polynomials $\mathfrak{S}_u(x)$ such that $u \in S_n$. These Schubert polynomials are uniquely expressed in terms of strict elementary symmetric monomials with fewer than n variables. Since Λ_n is a polynomial ring in the $e_{i,n}$ by the standard characterization of Λ_n as a polynomial ring in $e_1, \ldots, e_n$, Λ_n has a basis of monomials e_λ^n as λ ranges over all partitions with parts bounded by n. The multiplicative combination of these two bases is therefore a $\mathbb{Z}$-basis of $\mathcal{A}_n$ by freeness of the module. This is exactly the description of the monomials e_α^n for α an n-elementary weak composition. □

We can ask for transition formulas between the Schubert and the elementary basis. From elementary to Schubert, we may use Sottile's Pieri formula iteratively [31], which provides a positive formula for multiplying a Schubert polynomial by an elementary symmetric polynomial. That is,

$$e_\alpha^n = \sum_{1 \xrightarrow{1,2,\ldots,n}_\alpha u} \mathfrak{S}_u^n$$

For the reverse transition, we need a lemma. Recall the skew divided difference operators ∂_u^w for $u, w \in S_\infty$ defined originally in [19], though we use the notation of [28], by the Leibniz formula

$$\partial^w(pq) = \sum_u \partial^u(p)\partial_u^w(q)$$

where p and q are arbitrary polynomials and u is a permutation such that $u \leq w$ in Bruhat order. The formula below provides a formula for application of a skew divided difference operator of degree $-n$ to a polynomial of degree n, which yields an integer for integer-coefficient polynomials. Specifically, it is a formula for application of such an operator with $\ell(u, w) = n$ specifically to x_p^n for some $p > 0$.

Definition 2.4.2. For $u, w \in S_\infty$ such that $u \leq w$, we define $\mathcal{C}^{(p)}(u, w)$ to be the set of saturated chains C in the Bruhat interval $[u, w]$ of the form

$$u = u_0 \prec u_1 \prec \cdots \prec u_n = w$$

such that $u_{i-1}(p) \neq u_i(p)$ for all i. We also define $\sigma(C)$ to be the number of indices i for which the index $j \neq p$ such that $u_{i-1}(j) \neq u_i(j)$ satisfies $j < p$.

Lemma 2.4.3. *Let $n \geq 0$ be an integer, let $p > 0$ be an integer, and let $u, w \in S_\infty$ be permutations such that $\ell(u, w) = n$. Then we have that*

$$\partial_u^w(x_p^n) = \sum_{C \in \mathcal{C}^{(p)}(u,w)} (-1)^{\sigma(C)}.$$

Proof. Consider the case where $\ell(u,w) = 1$. Then there is a transposition t_{ij} such that $w = ut_{ij}$. Assume that $i < j$. If $p \notin \{i,j\}$, then $\partial_u^w(x_p) = 0$. If $p = i$, then $\partial_u^w(x_p) = \partial^{ij}(x_p) = 1$. If instead $j = p$, then $\partial_u^w(x_p) = \partial^{ij}(x_p) = -1$. In all cases, the result holds.

Now, consider the case where $\ell(u,w) = n > 1$. We have that

$$x_p^n = x_p \cdot x_p^{n-1}$$

Applying the Leibniz formula, we have

$$\partial_u^w(x_p^n) = \sum_{u \prec u'} \partial_u^{u'}(x_p)\partial_{u'}^w(x_p^{n-1})$$

By the above, $\partial_u^{u'}(x_p)$ is nonzero if and only if $u \prec u'$ is a cover in the Bruhat order such that $u(p) \neq u'(p)$. In this case, $\partial_u^{u'}(x_p)$ is either 1 or -1 depending on the number of indices $j < p$ such that $u(j) \neq u'(j)$, which is either 0 or 1, respectively. By induction,

$$\partial_{u'}^w(x_p^{n-1}) = \sum_{C' \in \mathcal{C}^{(p)}(u',w)} (-1)^{\sigma(C')}.$$

Hence,

$$\partial_u^{u'}(x_p)\partial_{u'}^w(x_p^{n-1}) = \begin{cases} -\displaystyle\sum_{C' \in \mathcal{C}^{(p)}(u',w)} (-1)^{\sigma(C')} & \text{if } u(j) \neq u'(j) \text{ for some } j < p \\ \displaystyle\sum_{C' \in \mathcal{C}^{(p)}(u',w)} (-1)^{\sigma(C')} & \text{otherwise,} \end{cases}$$

which, by definition, is

$$\sum_{C = C' \cup \{u\}, C' \in \mathcal{C}^{(p)}(u',w)} (-1)^{\sigma(C)}$$

Summing over all u' such that $u \prec u' \leq w$, we obtain the result. □

Using a minor sleight of hand, this affords us an explicit formula for transition of a Schubert polynomial into the elementary basis.

Definition 2.4.4. Let δ^u be the divided difference operator associated to u on the y variables for a given $u \in S_\infty$, and let

$$E_k(x; y_p) = \prod_{i=1}^{k} (x_i - y_p)$$

which is a factorial elementary symmetric polynomial, a special case of a double Schubert polynomial.

Lemma 2.4.5. *Let $u, w \in S_\infty$ satisfy $u \leq w$ and let $p > 0, k > 0$ be integers. Then we have*

$$\partial_u^w(E_k(x; -y_p)) = \sum_{C \in \mathcal{C}^{(p)}(u,w)} (-1)^{\sigma(C)} e_{k-\ell(u,w)}^k + \mathcal{O}(y_p)$$

Proof. This follows from the observation that

$$E_k(x; -y_p) = \sum_{i=0}^{k} e_{k-i}^k y_p^i$$

and Lemma 2.4.3. □

Definition 2.4.6. Suppose we want to express $\mathfrak{S}_w^n$ in the elementary basis. Let w_λ be a dominant permutation such that

$$\mathfrak{c}^*(w_\lambda) = \lambda = (n, n, \ldots, n, n-1, \ldots, 1)$$

and consider the double Schubert polynomial

$$\mathfrak{S}_{w_\lambda}(x; -y) = \prod_i E_{\lambda_i}(x; -y_i)$$

(this equation for $\mathfrak{S}_{w_\lambda}$ follows from [19, (6.14)]).

We have that

$$\mathfrak{S}_w(x; y) = \delta^{w\lambda^{-1}} \mathfrak{S}_{w_\lambda}(x; y)$$

We may apply the divided difference $\delta^{ww_\lambda^{-1}}$ to the product $\prod_i E_{\lambda_i}(x;-y_i)$ using the Leibniz formula and set $y_i = 0$ for all i, which gives us, by Lemma 2.4.5,

$$\sum_{\substack{P=(C_1,\ldots,C_{\ell(\lambda)})\\ C_i\in\mathcal{C}^{(i)}(u_{i-1},u_i)}} (-1)^{\sigma(C_1)+\cdots+\sigma(C_{\ell(\lambda)})} e^{\lambda_1}_{\lambda_1-|C_1|} e^{\lambda_2}_{\lambda_2-|C_2|}\cdots e^{\lambda_{\ell(\lambda)}}_{\lambda_{\ell(\lambda)}-|C_{\ell(\lambda)}|}$$

where $u_0 \le u_1 \le \cdots \le u_{\ell(\lambda)}$ is a saturated chain. Define, for convenience,

$$\mathrm{Path}_\lambda(w) = \{P = (C_1,\ldots,C_{\ell(\lambda)}) \mid C_i \in \mathcal{C}^{(i)}(u_{i-1},u_i), u_0 = 1, u_{\ell(\lambda)} = ww_\lambda^{-1}\}$$

and for $P \in \mathrm{Path}_\lambda(w)$ define an integer

$$\sigma(P) = \sum_{i=1}^{\ell(\lambda)} \sigma(C_i)$$

Furthermore, define a weak composition $\alpha(P)$ by

$$\alpha(P) = (\lambda_{\ell(\lambda)+1-i} - |C_{\ell(\lambda)+1-i}|)_{i=1}^{\ell(\lambda)}$$

Then we define

$$E_w^{\alpha;n} = \sum_{\substack{P\in\mathrm{Path}_\lambda(w)\\ \alpha(P)=\alpha}} (-1)^{|\lambda|-\ell(w)+\sigma(P)}$$

Theorem 2.2 (Transition to the elementary basis)**.** *For each w, we have*

$$\mathfrak{S}_w^n = \sum_\alpha E_w^{\alpha;n} e_\alpha^n$$

These numbers are stable for fixed w as soon as the number of variables is at least as large as the value of n such that $w \in S_n$, and are well-studied but poorly understood.

Example 2.4.7. Let $n = 5$ and let w be the permutation such that $\mathfrak{c}(w) = (0,2,0,2,3)$. Then

$$\begin{aligned}\mathfrak{S}_w^n &= e_{(0,0,0,0,5,1,1)} - e_{(0,0,0,0,4,2,1)} + e_{(0,0,0,0,4,3)} - e_{(0,1,0,0,4,1,1)} + e_{(0,0,1,0,3,2,1)}\\ &\quad + e_{(0,1,0,0,5,1)} + e_{(0,1,0,0,3,3)}\end{aligned}$$

Remark. This transition formula from the Schubert basis to the elementary basis can be used to obtain an extremely efficient formula for multiplication of Schubert polynomials. If we want to compute $\mathfrak{S}_u^n\mathfrak{S}_v^n$, we can first express $\mathfrak{S}_v^n$ in the elementary basis using Theorem 2.2. Then we can multiply by $\mathfrak{S}_v^n$ by iterating the Pieri formula. This is the method used in the python package `schubmult` written by the author [30]. In fact, there is also a version for double Schubert polynomials, quantum Schubert polynomials, and quantum double Schubert polynomials, which we will not go into here but are implemented in the same package.

3. The dual algebra

We examine the graded dual algebra $\mathcal{D}$ more closely now. This is a graded ring generated by countably many elements that we denote by $[i]$, where i is a nonnegative integer. The product, as mentioned previously, is concatenation of sequences. Thus

$$[a_1\cdots a_p][b_1\cdots b_q] = [a_1\cdots a_p b_1\cdots b_q]$$

It is not hard to see that $\mathcal{D}$ is a free algebra on these generators. Thus the set of sequences $[a_1\cdots a_n]$ forms a $\mathbb{Z}$-basis for $\mathcal{D}$. We may realize this as the dual of $\mathcal{A}$ by declaring that

$$\langle [a_1\cdots a_n], x_1^{b_1}\cdots x_n^{b_n}\rangle = \prod_i \delta_{a_i,b_i}$$

We have a $\mathbb{Z}\times\mathbb{Z}$ grading such that

$$\deg([a_1\cdots a_n]) = (n, -a_1 - a_2 \cdots - a_n)$$

The set of elements such that $\deg(a) = (n,-)$ is $\mathcal{D}_n$, dual as a graded module to $\mathcal{A}_n$.

3.1. **The dual Schubert basis.** There is a basis Ξ_u^n dual to the Schubert basis for $\mathcal{D}_n$. Specifically, with the unique pairing $\langle -, - \rangle : \mathcal{D} \times \mathcal{A} \to \mathbb{Z}$ such that

$$\langle \alpha, x_\beta \rangle = \delta_{\alpha\beta}$$

we define Ξ_u^n to be the unique basis of $\mathcal{D}_n$ such that

$$\langle \Xi_u^n, \mathfrak{S}_v^n \rangle = \delta_{uv}$$

We characterize it with an explicit formula.

Theorem 3.1. *For each permutation u and integer n with $\mathfrak{m}(u) \leq n$ we have the equation*

$$\Xi_u^n = \sum_{\ell(\alpha)=n} E_{u\mu^{-1}}^{\mathfrak{c}(\mu)-\alpha;n} \alpha$$

where μ is any strict dominant permutation such that $0 \neq \mathfrak{c}_n(\mu) \geq \ell(u)$ and $\mathfrak{c}_{n+1}(\mu) = 0$.

Proof. By definition, the coefficient of α in Ξ_u^n is the coefficient of $\mathfrak{S}_u^n$ in x_α. This can be derived from the Cauchy formula for double Schubert polynomials. Note that for any permutation μ as laid out in the statement of the theorem, $\ell(u\mu^{-1}) = \ell(\mu) - \ell(u)$. Thus,

$$\partial_y^{u\mu^{-1}} \mathfrak{S}_\mu(x; -y) = \mathfrak{S}_u(x; -y)$$

We have that

$$\mathfrak{S}_\mu(x; -y) = \sum_u \mathfrak{S}_u(x)\, \mathfrak{S}_{u\mu^{-1}}(y)$$

Expressing the second factor in the $e_{\alpha,\lambda}(y)$ basis, we have

$$\mathfrak{S}_\mu(x; -y) = \sum_{u,\alpha} \mathfrak{S}_u(x) E_{u\mu^{-1}}^{\mathfrak{c}(\mu)-\alpha;n} e_{\mathfrak{c}(\mu)-\alpha,\mathfrak{c}(\mu)}(y)$$

An alternative expression for $\mathfrak{S}_\mu(x; -y)$ is

$$\mathfrak{S}_\mu(x; -y) = \sum_\alpha x_\alpha e_{\mathfrak{c}(\mu)-\alpha,\mathfrak{c}(\mu)}(y)$$

from which we see that the coefficient is correct. □

This is not stable for fixed u as n increases, and this is expected.

Example 3.1.1. Suppose $w = [1, 4, 5, 2, 3]$. Then

$$\Xi_w^3 = [022] - [013]$$

If we chose $[1, 3, 5, 7, 2, 4, 6]$ instead, we have

$$\Xi_{1357246} = [0015] - [0033] - [0114] + [0123]$$

For $w = [4, 2, 7, 1, 3, 5, 6]$,

$$\Xi_{4271356}^3 = -[026] + [035] + [125] - [134] + [206] - [215] - [305] + [314]$$

Thus the product structure of $\mathcal{D}$ encodes splitting of the Schubert polynomial into two sets of variables. In particular,

Lemma 3.1.2. *Let $a \geq 0$ be an integer and $w \in S_\infty$. Then we have*

$$[a] \cdot \Xi_w^n = \sum_{\substack{w' \overset{1}{\searrow} w \\ \ell(w,w')=a}} \Xi_{w'}^{n+1}$$

Proof. This follows from Proposition 2.1.4 with $i = 1$. □

3.2. **The coproduct.**

4. The ring of bounded RC graphs

4.1. The module of bounded RC graphs.

Definition 4.1.1. Let $R \subseteq \mathbb{P} \times \mathbb{P}$ be a finite set. To each such set we associate an element w_R of S_∞ as follows. Given a pair (i,j) where $i,j > 0$ are integers, define $s(i,j) = s_{i+j-1}$. Totally order the grid such that $(i,j) < (a,b)$ if and only if $i < a$ or $i = a$ and $j > b$ (in other words, lexicographical order, except that the order on the second coordinate is reversed). By this ordering, index R as $r_1, r_2, \dots, r_m$ in increasing order. Then

$$w_R = s_{r_1} \cdots s_{r_m}$$

If $\ell(w_R) = m$, then we say that R is an *RC-graph.*

A *bounded RC-graph* is an RC-graph R together with a specified number of rows $n \geq \max\{i : (i,j) \in R \text{ for some } j\}$ such that w_R has no right descent larger than n. The definition is as an ordered pair (R,n), but it will be far more convenient to abuse notation and consider the number of rows a property of R itself. To denote the number of rows, we define the *height* of R to be $\mathtt{ht}(R) = n$. A bounded RC graph has an associated weight vector $\mathtt{wt}(R)$ where $\mathtt{wt}_i(R)$ is the number of elements in row i.

To a bounded RC graph R with $\mathtt{ht}(R) \geq 1$, we define the *trim* operation $\mathtt{trim}(R)$ to be the bounded RC graph with height $\mathtt{ht}(R) - 1$ and underlying set $\{(i-1, j-1) : (i,j) \in R, i \geq 2\}$. We also define

$$\uparrow^m R = \{(i+m, j) | (i,j) \in R\}$$

Let $\mathcal{BRC}$ be the free abelian group spanned by all bounded RC graphs. If $\mathcal{BRC}_n$ is the subgroup spanned by all bounded RC graphs R with $\mathtt{ht}(R) = n$, then we have a grading

$$\mathcal{BRC} = \bigoplus_{n=0}^{\infty} \mathcal{BRC}_n$$

There is an evident evaluation map $\phi : \mathcal{BRC} \to \mathcal{A}$ defined on basis elements as

$$\phi(R) = x_{\mathtt{wt}(R)}$$

which is a surjective homomorphism of graded modules. There is also a map $\alpha : \mathcal{BRC} \to \mathcal{D}$ defined by

$$\alpha(R) = \Xi_{w_R}^{\mathtt{ht}(R)}$$

which is also a surjective homomorphism of graded modules. In addition, there is $\omega : \mathcal{BRC} \to \mathcal{D}$ defined by

$$\omega(R) = [\mathtt{wt}(R)]$$

which is similarly surjective.

We can define elements $\mathcal{S}_w(n)$ as

$$\mathcal{S}_w(n) = \sum_{\substack{w_R = w \\ \mathtt{ht}(R) = n}} R$$

For a generating element $[a]$ of $\mathcal{D}$ and a bounded RC graph R, we define

$$[a] \cdot R = \sum_{\substack{\mathtt{trim}(R') = R \\ \mathtt{wt}_1(R') = a}} R'$$

which is an element of $\mathcal{BRC}$. By virtue of the fact that $\mathcal{D}$ is a free algebra generated by these elements, it is nearly a trivial observation that this is a left module action on $\mathcal{BRC}$. The consequences of this, however, are not at all trivial.

Lemma 4.1.2 ([3, Corollary 3.11])**.** *We have that the set $\{w' \mid w \overset{1}{\searrow} w'\}$ is equal to the set of permutations w' such that there exists a permutation $v = s_{a_1} \cdots s_{a_m}$ for some integers $a_1 > a_2 > \cdots > a_m \geq 1$ such that $w = v \uparrow w'$.*

Lemma 4.1.3. *Let v be a permutation. If $v \overset{1}{\searrow} v'$, let $a_1, \dots, a_k$ be the elements of $Q_1(v', v)$ in decreasing order. Then*

$$v = s_{a_1} \cdots s_{a_k} \uparrow v'$$

Proof. Suppose $v \in S_n$. We have by definition that there is a sequence of integers $b_1, \ldots, b_p$, all distinct and greater than 1, such that

$$vt_{1,b_1} \cdots t_{1,b_p}(1) = n+1$$

and

$$vt_{1,b_1} \cdots t_{1,b_p}(i+1) = v'(i)$$

for all $i < n-1$. This means that

$$vt_{1,b_1} \cdots t_{1,b_p} = s_n s_{n-1} \cdots s_1 {\uparrow} v'$$

This is because if $v'' = {\uparrow} v'$, then

$$v''(1) = 1$$

and

$$v''(i) = v(i-1) + 1$$

The cycle $s_n \cdots s_1$ sends $1 \mapsto n+1$ and $i \mapsto i-1$ if $1 < i \leq n+1$, hence

$$vt_{1,b_1} \cdots t_{1,b_p} = s_n s_{n-1} \cdots s_1 {\uparrow} v'$$

In particular,

$$v = s_n s_{n-1} \cdots s_1 {\uparrow} v' t_{1,b_p} \cdots t_{1,b_1}$$

This results in the factorization

$$v = s_n \cdots s_1 t_{1,v'(b_p-1)+1} \cdots t_{1,v'(b_1-1)+1} {\uparrow} v'$$

The value $v'(b_j - 1)$ necessarily decreases as j decreases, since applying the corresponding reflection in the reverse order strictly increases the length with each application. Consequently, multiplying $s_n \cdots s_1$ on the right by these reflections removes the simple reflections $s_{v'(b_p-1)}, \ldots, s_{v'(b_1-1)}$. The indices removed are precisely the complement of the elements of $Q_1(v', v)$, hence the elements of $Q_1(v', v)$ in decreasing order are what remain, as desired. □

Definition 4.1.4. For a set of positive integers $A = \{a_1, \ldots, a_m\}$ with $a_1 > a_2 > \cdots > a_m \geq 1$ and a positive integer i, define

$$\mathrm{row}_i(A) = \{(i, a_j) \mid a_j \in A\}$$

Theorem 4.1. *Let $a \geq 0$ be an integer and let $R \in \mathcal{BRC}$. Then*

$$[a] \cdot R = \sum_{\substack{w' \overset{1}{\searrow} w_R \\ \ell(w_R, w') = a}} (\mathrm{row}_1(Q_1(w_R, w')) \cup {\uparrow} R)$$

where the right side denotes bounded RC graphs with $\mathtt{ht}(R) + 1$ *rows.*

Proof. If $\mathtt{trim}(R') = R$, then by definition $w_{R'} = s_{a_1} \cdots s_{a_m} {\uparrow} w_R$. It follows by Lemma 4.1.2 then $w_{R'} \overset{1}{\searrow} w_R$. By Lemma 4.1.3, the integers $a_1, \ldots, a_m$ are precisely the elements of $Q_1(w_R, w_{R'})$ in decreasing order. This establishes the result. □

Lemma 4.1.5. *Let $a \geq 0$ be an integer and let $w \in S_\infty$. For any valid n, we have*

$$[a] \cdot \mathcal{S}_w(n) = \sum_{\substack{w \overset{1}{\searrow} w' \\ \ell(w, w') = a}} \mathcal{S}_{w'}(n+1)$$

Proof. Theorem 4.1 establishes an explicit bijection between $[a] \cdot R$ and the set of bounded RC graphs R' such that $w_{R'} \overset{1}{\searrow} w_R$ and $\ell(w_R, w_{R'}) = a$. The result follows by summing over all R such that $w_R = w$ and $\mathtt{ht}(R) = n$. □

Let t be an indeterminate commuting with all elements of $\mathcal{D}$. Write

$$\mathfrak{S}_{\mathcal{D}}(t) = \sum_{a=0}^{\infty} [a] t^a$$

Define

$$\mathfrak{S}_{\mathcal{D}}(x_1, x_2, \ldots, x_n) = \mathfrak{S}_{\mathcal{D}}(x_1)\, \mathfrak{S}_{\mathcal{D}}(x_2) \cdots \mathfrak{S}_{\mathcal{D}}(x_n)$$

Theorem 4.2. *Let $\emptyset \in \mathcal{BRC}$ denote the empty bounded RC graph with $\mathtt{ht}(\emptyset) = 0$. Then*

$$\mathfrak{S}_{\mathcal{D}}(x_1, \ldots, x_n) \cdot \emptyset = \sum_{\substack{w \in S_\infty \\ \mathfrak{m} w \leq n}} \mathfrak{S}_w(x_1, \ldots, x_n) \mathcal{S}_w(n)$$

Proof. The result for $n = 1$ is Lemma 4.1.5 together with Theorem 4.1. The general result follows by induction on n. Consider the inductive hypothesis

$$\mathfrak{S}_{\mathcal{D}}(x_2, \ldots, x_n) \cdot \emptyset = \sum_{w \in S_\infty} \mathfrak{S}_w(x_2, \ldots, x_n) \mathcal{S}_w(n-1)$$

Then applying $\mathfrak{S}_{\mathcal{D}}(x_1)$ to both sides, we have

$$\mathfrak{S}_{\mathcal{D}}(x_1)\, \mathfrak{S}_{\mathcal{D}}(x_2, \ldots, x_n) \cdot \emptyset = \sum_{w \in S_\infty} \mathfrak{S}_w(x_2, \ldots, x_n)\, \mathfrak{S}_{\mathcal{D}}(x_1) \mathcal{S}_w(n-1)$$

which is equal to

$$\sum_{a=0}^{\infty} \sum_{w \in S_\infty} x_1^a\, \mathfrak{S}_w(x_2, \ldots, x_n)[a] \mathcal{S}_w(n-1)$$

Applying Lemma 3.1.2 to each term, we have

$$\sum_{a=0}^{\infty} \sum_{w \in S_\infty} x_1^a\, \mathfrak{S}_w(x_2, \ldots, x_n) \quad \sum_{\substack{w' \overset{1}{\searrow} w \\ \ell(w, w') = a}} \mathcal{S}_{w'}(n)$$

Bringing the polynomial inside the inner sum, this is

$$\sum_{w' \in S_\infty} \left(\sum_{w : w' \overset{1}{\searrow} w} x_1^{\ell(w,w')}\, \mathfrak{S}_w(x_2, \ldots, x_n) \right) \mathcal{S}_{w'}(n)$$

which simplifies to

$$\sum_{w' \in S_\infty} \mathfrak{S}_{w'}(x_1, \ldots, x_n) \mathcal{S}_{w'}(n)$$

as desired. □

Corollary 4.1.6. *For each $w \in S_\infty$ and valid n, we have*

$$\phi(\mathcal{S}_w(n)) = \mathfrak{S}_w^n(x_1, \ldots, x_n)$$

4.2. **The pipe dream visualization and roots.** Given a pair of positive integers i, j and an RC graph R, define

$$R[i,j] = \{(a,b) \in R \mid (a,b) > (i,j)\}$$

Given an RC graph R and a pair of positive integers i, j, we define an ordered pair

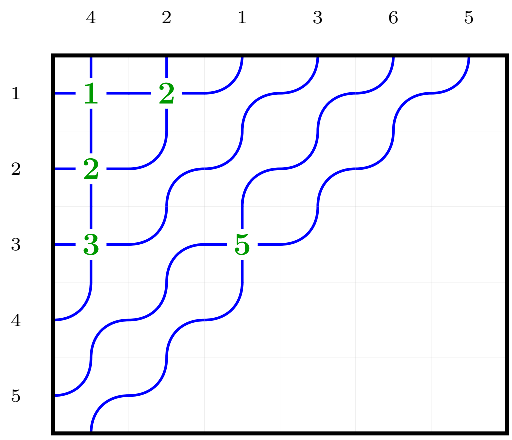
$$\mathtt{rt}_R(i,j) = (w_{R[i,j]}^{-1}(i+j-1), w_{R[i,j]}^{-1}(i+j))$$

There is a common visualization of RC graphs as pipe dreams. In this visualization, we draw an infinite grid in the first quadrant, and in each position (i, j) we draw either a crossing (if $(i, j) \in R$) or an elbow (if $(i, j) \notin R$). Then we draw pipes entering from the left edge of the grid, with the pipe entering at row i labeled i. The pipes travel through the grid, turning at elbows and crossing at crossings, and exit at the top of the grid, which is labeled with the same number.

A modification to this common visualization that we use has the following additional features:

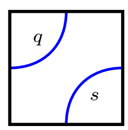
- We write the index of the simple reflections corresponding to the crossings in the grid area itself. Thus, at position (i, j) we write the number $i + j - 1$ if there is a crossing at that position.
- For a bounded RC graph R with $\mathtt{ht}(R) = n$, we only draw the first n pipes entering from the left side of the grid and clip features outside of the first n rows. The pipes exiting at the top are still labeled since the width is not limited.
- *Note:* The labels at the top are not of the permutation w_R, but rather of the permutation w_R^{-1}.

FIGURE 1. The pipe dream visualization of the bounded RC graph R with $\mathtt{ht}(R) = 5$ where $R = \{(1,1),(1,2),(2,1),(3,1),(3,3)\}$



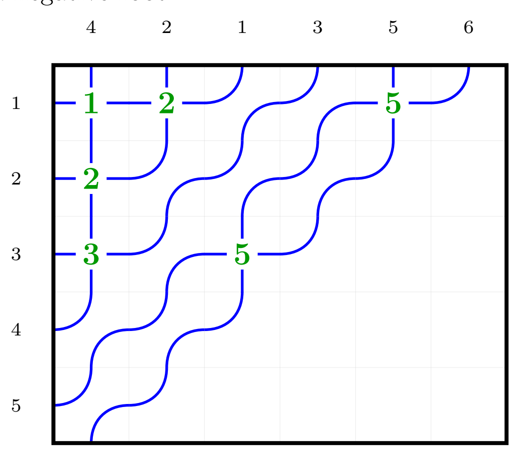
See Figure 1 for an example of this visualization.

The main benefit of this is that visualizing $\mathtt{rt}_R(i,j)$ is easy. The pipes that pass through position (i,j) are labeled s and q for some $s, q > 0$. For a positive root in an unoccupied square, we will have the following labeling, where $q < s$:



We observe that in the above RC graph, the position $(1,5)$ does not have this configuration. Placing a crossing there would create a negative root, and the pipes would cross twice. *Note that this results in a collection of ordered pairs that is not an RC graph, if such a crossing exists.* For a valid RC graphs, only positive roots occur as crossings, and this happens if and only if pipes cross at most once.

FIGURE 2. An invalid set of crossings causing pipes to cross more than once, caused by inserting a crossing at a negative root



4.3. **Zeroing out the last row.** We proceed now to define a product on $\mathcal{BRC}$ turning it into a ring. To do this, we need to be able to define a function $\mathcal{Z} : \mathcal{BRC} \to \mathcal{BRC}$ trimming empty rows from the bottom instead of from the top. This is far more complicated.

See Figure 3 for an example of the zeroing operation (Algorithm 3).

Algorithm 1 Basic monk insertion $i \overset{k}{\rightsquigarrow} R$

1: **Input:** An RC graph R, parameter k, and an integer i with $1 \le i \le k$.
2: **Output:** A modified RC graph R' with $\mathtt{wt}_j(R') = \mathtt{wt}_j(R)$ for $j \ne i$ and $\mathtt{wt}_i(R') = \mathtt{wt}_i(R) + 1$ such that $w_R \xrightarrow{k} w_{R'}$.
3: Find leftmost position $(i, j) \notin R$ in row i such that if $\mathtt{rt}_R(i, j) = (a, b)$, then $a \le k < b$.
4: $R \leftarrow R \cup \{(i, j)\}$
5: **if** there exists $(i', j') \in R$ with $\mathtt{rt}_R(i', j') \in \Phi^-$ **then**
6: $\quad R \leftarrow R \setminus \{(i', j')\}$
7: $\quad$ Return to step 3 substituting $i = i'$.
8: **end if**
9: **return** R

Algorithm 2 Pseudo-Pieri insertion $I \overset{k}{\rightsquigarrow} R$

1: **Input:** An RC graph R, parameter k, and sequence $I = \{i_1, \ldots, i_m\}$ with $k \ge i_1 \ge i_2 \ge \cdots \ge i_m \ge 1$.
2: **Output:** A modified RC graph R' with $\mathtt{wt}_i(R') = \mathtt{wt}_i(R)$ for $i \notin I$ and $\mathtt{wt}_i(R') = \mathtt{wt}_i(R) + \#\{j \mid i_j = i\}$ for $i \in I$ such that $w_R \overset{k}{\Longrightarrow} w_{R'}$.
3: **Initialize:** Let $L \leftarrow [\,]$ (empty list of pairs (a, b) where $a \le k < b$), and $U \leftarrow [\,]$ (empty list of positions).
4: **for** each $i \in (i_1, \ldots, i_m)$ **do**
5: $\quad$ Find leftmost position $(i, j) \notin R$ in row i satisfying $j > j'$ for all $(i, j') \in U$ such that either:
6: $\quad\quad$ a) $\mathtt{rt}_R(i, j) = (s, q)$ where $s \le k < q$ and $q \notin \{b \mid (a, b) \in L\}$.
7: $\quad\quad$ b) $\mathtt{rt}_R(i, j) = (b_r, q)$ where $q > k$, $b_r < q$, and $q \notin \{b \mid (a, b) \in L\}$.
8: $\quad R \leftarrow R \cup \{(i, j)\}$ and $U \leftarrow U \cup \{(i, j)\}$
9: $\quad$ Update L by adding (s, q) or replacing (a_r, b_r) with (a_r, q) and (a_r, b_r).
10: $\quad$ **if** there exists $(i', j') \in R$ with $\mathtt{rt}_R(i', j') \in \Phi^-$ **then**
11: $\quad\quad R \leftarrow R \setminus \{(i', j')\}$
12: $\quad\quad$ Let $\mathtt{rt}_R(i', j') = (q, s)$
13: $\quad\quad$ **if** $(s, q) \in L$ **then**
14: $\quad\quad\quad$ Remove (s, q) from L
15: $\quad\quad$ **else**
16: $\quad\quad\quad$ Remove (a_r, q) from L, where $(a_r, q), (a_r, s) \in L$
17: $\quad\quad$ **end if**
18: $\quad\quad$ Return to step 5 to insert i'.
19: $\quad$ **end if**
20: **end for**
21: **return** R

Note that Algorithm 2 is *not* a realization of Sottile's Pieri formula for complete symmetric polynomials, despite preserving the relevant Bruhat relation. This is because the map

$$\left(\!\!\binom{k}{p}\!\!\right) \times \mathcal{RC}(w) \to \mathcal{RC}(n)$$

given by

$$(I, R) \mapsto I \overset{k}{\rightsquigarrow} R$$

need not be injective.

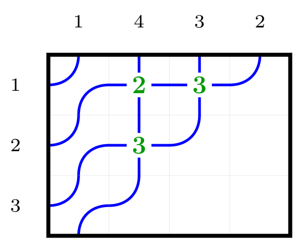
Definition 4.3.1. Let R be an RC graph with $\mathtt{ht}(R) = n$ such that row n is empty. We define $\mathcal{Z}(R)$ to be the output of Algorithm 3, an RC graph with $n-1$ rows, on input R.

Lemma 4.3.2. *Given an RC graph R, an integer k, and $k \ge i_1 \ge \cdots \ge i_m \ge 1$, Algorithm 2 produces a valid RC graph R' satisfying*

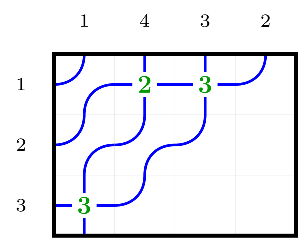
$$\mathtt{wt}_i(R') = \mathtt{wt}_i(R) + \#\{j \mid i_j = i\}$$

Algorithm 3 Map $\mathcal{Z}(R)$ zeroing out last row

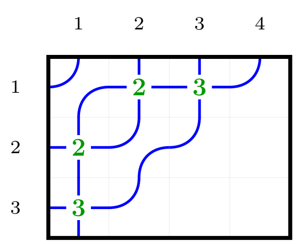
Input: A bounded RC graph R with $\mathtt{ht}(R) = n$ and row n empty.
Output: A bounded RC graph R' with $\mathtt{ht}(R') = n-1$, $|R'| = |R|$ (same number of crossings), and $w_R \overset{n}{\searrow} w_{R'}$.
if R with last row removed is a valid bounded RC graph **then**
    return R with last row removed
else
    Let $R_0 \leftarrow R$.
    Find the maximal p and sequence $\{(i_m, j_m)\}_{m=1}^{p}$ such that:
      $\mathtt{rt}_{R_{m-1}}(i_m, j_m) = (n+m-1, n+m)$ for each m, where $R_m = R_{m-1} \setminus \{(i_m, j_m)\}$ for each $m \geq 1$.
    $R^- \leftarrow R \setminus \{(i_1, j_1), \ldots, (i_p, j_p)\}$.
    Let $I \leftarrow (i_1, i_2, \ldots, i_p)$.
    $D \leftarrow R^- \cup \{(n,1), (n,2), \ldots, (n,p)\}$.
    $D' \leftarrow I \overset{n-1}{\rightsquigarrow} D$
    $R' \leftarrow \{(i,j) \in D' \mid i < n\}$ as a bounded RC graph with $n-1$ rows.
    return R'.
end if



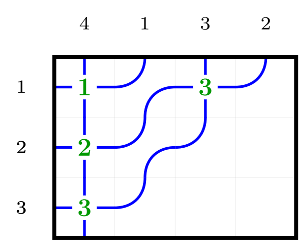
(a) Initial RC graph R



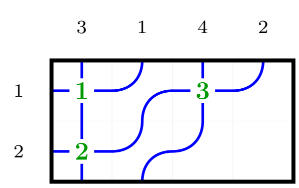
(b) After deleting $(2,2)$ with $\mathtt{rt}_R(2,2) = \alpha_3$ and moving to row 3



(c) After insertion with $i_1 = 2$



(d) After rectification



(e) Final result $\mathcal{Z}(R) = (R', 2)$ after removing row 3

FIGURE 3. Step-by-step computation of $\mathcal{Z}(R)$ for $R = \{(1,2), (1,3), (2,2)\}$.

and

$$w_R \stackrel{k}{\Longrightarrow} w_{R'}$$

Proof. Set $R_0 = R$ to be the initial RC graph. We claim that given R and L at the start of iteration p of the main loop, we have that R is a valid RC graph satisfying

$$\mathtt{wt}_i(R) = \mathtt{wt}_i(R_0) + \#\{j \le p-1 \mid i_j = i\}$$

and

$$w_R = w_{R_0} t_{a_1 b_1} \cdots t_{a_{p-1} b_{p-1}}$$

where $L = [(a_1, b_1), \ldots, (a_p, b_p)]$. This is clear for $p = 1$. For larger p, suppose we are inserting at row i_p. If we are in case (a) of Step 5, then adding (i_p, j) to R adds the root $t_s - t_q$ to the inversion set of w_R, and multiplying by t_{sq} gives the desired result. If we are in case (b) of Step 5, then adding (i_p, j) to R results in multiplication by $t_{b_r q}$. This commutes past the other reflections to meet $t_{a_r b_r}$. We thus have the situation

$$t_{a_r b_r} t_{b_r q} = t_{a_r q} t_{a_r b_r}$$

This ensures that the product remains as desired if the RC graph is valid. However, the condition that the RC graph R be valid is not guaranteed after this step, so we must rectify. Suppose we must rectify at row $i-1$. We find a negative root $(i-1, j')$ which must have been created by the insertion, and hence by the strong exchange property equal to $\mathtt{rt}_R(i,j)$ that was inserted. Removing this root (from both the graph and from L) returns us to the original permutation at the beginning of the iteration, and performing the insert at row $i-1$ adds a new positive root, giving the desired result on the permutation. The weight is unchanged since we removed and added one crossing in row $i-1$. If the algorithm ends here, we have instead multiplied by the new reflection obtained in the ultimate insert step. Otherwise, repeating this process for all necessary rectifications completes the proof of the claim. The claim iterated to $p = m+1$ yields the desired result, and the method of updating L ensures that the b_j are all distinct, so that $w_R \stackrel{k}{\Longrightarrow} w_{R'}$. □

Lemma 4.3.3. *Given a bounded RC graph R with $\mathtt{ht}(R) = n$ such that row n is empty, Algorithm 3 produces a valid bounded RC graph R' with $\mathtt{ht}(R') = n-1$ satisfying*

$$\mathtt{wt}(R') = \mathtt{wt}(R)$$

Proof. Stripping out the crossings at roots $(n, n+1), (n+1, n+2), \ldots, (n+p-1, n+p)$ removes exactly one crossing from each of rows $i_1, i_2, \ldots, i_p$, and moving them to the last row to obtain R_1 ensures that $w_R = w_{R_1}$. By construction, the portion of the RC graph in rows with index less than n is valid (meaning its max descent is at most $n-1$). Applying Algorithm 2 to insert at rows $i_1, i_2, \ldots, i_p$ adds different crossings to preserve the number of elements in each row without adding descents past index n in the permutation, by Lemma 4.3.2. Removing row n then yields a valid bounded RC graph R' with $|R'| = n-1$ and the desired properties. □

Theorem 4.3. *Let w be a permutation with last descent at most n. Let $\mathcal{RC}(w,n)^0$ be the set of bounded RC graphs R with $\mathtt{ht}(R) = n$ such that $w_R = w$ and row n is empty. Then*

$$\mathcal{Z} : \mathcal{RC}(w,n)^0 \to \bigcup_{w \overset{n}{\searrow} w'} \mathcal{RC}(w', n-1)$$

is a weight-preserving bijection.

To prove Theorem 4.3 for $n = 2$, we may characterize the bounded RC graphs R with $|R| = 2$ such that row 2 is empty and $\mathtt{m}(w_R) = 2$ as those for permutations $w = s_k s_{k-1} \cdots s_2$ for some $k > 1$. Algorithm 3 in this case consists moves the entirety of the crossings in row 1 to row 2. The procedure then inserts $k-1$ crossings into row 1 in order from left to right, which must have roots $t_q - t_s$ such that $q = 1$. Thus R' is precisely $\{(1,1), (1,2), (1,3), \ldots, (1, k-1)\}$, which is the unique bounded RC graph for the permutation $w' = s_{k-1} s_{k-2} \cdots s_1$ with one row. This establishes the base case.

For the induction step, we require the following lemmas.

Lemma 4.3.4. *Let R be a bounded RC graph with $\mathtt{ht}(R) = n \ge 2$ such that row n is empty. Then if $\mathcal{Z}(R) = R'$, we have*

$$w_R \overset{n}{\searrow} w_{R'}$$

Proof. Let $w = w_R$ and suppose $\mathfrak{c}_n(w) = m$. By construction, the second to last step of Algorithm 3 yields a bounded RC graph $\widetilde{R}$ with $|\widetilde{R}| = n$ such that if $\widetilde{w} = w_{\widetilde{R}}$, then

$$\widetilde{w}(n) > \widetilde{w}(n+1) < \widetilde{w}(n+2) < \cdots < \widetilde{w}(n+m)$$

and

$$w \xRightarrow{n-1} \widetilde{w}$$

via reflections t_{ab} such that $n \leq b \leq n+m$. Let $w' = w_{R'}$. We also have

$$\widetilde{w} \overset{n}{\Rightarrow} \varphi_{n,N}(w')$$

via only reflections t_{nb} such that $b > n+m$. We claim that

$$w \overset{n}{\Rightarrow} \varphi_{n,N}(w')$$

This can only fail if $\widetilde{w}(n) \neq w(n)$ by the observations above. However, the pipe labeled n, by virtue of the fact that we have $(n,1),\ldots,(n,m)$ populated, will always lie to the right of pipes $n+1, n+2, \ldots, n+m$ in the wiring diagram for $\widetilde{w}$ or any intermediate stage. Inserting into any row that contained a crossing $(n, n+p)$, there must be a valid empty space to the left of the pipe labeled n since we have deleted these crossings. By virtue of the fact that the leftmost is always chosen, no reflection (i,n) will ever be inserted. This ensures that $\widetilde{w}(n) = w(n)$, as desired. □

Lemma 4.3.5. *Let (R,n) be a bounded RC graph with $n \geq 2$ such that row n is empty. Then*

$$\mathcal{Z}(\mathtt{trim}(R)) = \mathtt{trim}(\mathcal{Z}(R))$$

Proof. This is almost a trivial observation. In terms of the word of R, $\mathtt{trim}(R,n)$ preserves a suffix. Removing all of the initial roots before trimming is therefore the same as removing the initial roots that still remain after trimming, by the exchange property. Afterwards, in perfoming the insertion algorithm, there is no dependency on rows with a lower index, the modification only proceeds downward in row number. Therefore, trimming the first row at any point only stops the process earlier, and does not change rows with higher index than 1 in the outcome. □

Lemma 4.3.6. *Let w be a permutation with last descent at most n. Let $\mathcal{RC}(w,n)^0$ be the set of bounded RC graphs R with $\mathtt{ht}(R) = n$ such that $w_R = w$ and row n is empty. Then*

$$\mathcal{Z} : \mathcal{RC}(w,n)^0 \to \mathcal{RC}(n-1)$$

is injective.

Proof. If $n = 2$ this is clear. Suppose now that $n > 2$ and that the result holds for $n-1$. Let $(R,n) \in \mathcal{RC}(w,n)^0$. If $(R',n) \in \mathcal{RC}(w,n)^0$ is another bounded RC graph such that

$$\mathcal{Z}(R) = \mathcal{Z}(R')$$

then by Lemma 4.3.5, we have

$$\mathcal{Z}(\mathtt{trim}(R)) = \mathcal{Z}(\mathtt{trim}(R'))$$

hence R and R' agree on all rows except possibly row 1 by the inductive hypothesis. Since $w_R = w_{R'}$, it follows that $R = R'$, establishing injectivity. □

Proof of Theorem 4.3. By the previous lemma, it suffices to show that $\mathcal{Z}$ is surjective. However, this follows from the pigeonhole principle and the well-known transition formula for Schubert polynomials. □

We may extend $\mathcal{Z}$ to an endomorphism $\mathcal{BRC} \to \mathcal{BRC}$ by defining $\mathcal{Z}(R) = 0$ if the last row of R is not empty. Then we have the following result.

Theorem 4.4. *Let w be a permutation with last descent at most n. Then*

$$\mathcal{Z}(\mathcal{S}_w(n)) = \sum_{\substack{w \overset{n}{\searrow} w' \\ \ell(w) = \ell(w')}} \mathcal{S}_{w'}(n-1)$$

4.4. Definition of the ring product.

Definition 4.4.1. Suppose we have two bounded RC graphs R_1 with $\mathtt{ht}(R_1) = m$ and R_2 with $\mathtt{ht}(R_2) = n$. The product of these is defined by

$$R_1 \diamond R_2 = \sum_{\substack{\mathtt{clip}^m(R')=R_1 \\ \mathtt{trim}^m(R')=R_2}} R'$$

where R' ranges over bounded RC graphs with $\mathtt{ht}(R') = m + n$.

Definition 4.4.2. Define a function $A_p : \mathcal{RC}_n \to \mathcal{RC}_p \times \mathcal{RC}_{n-p} \times S_\infty$ as follows:

$$A_p(R) = (\mathtt{clip}^p(R), \mathtt{trim}^p(R), w_R)$$

Lemma 4.4.3. *The function A_p is injective.*

Proof. This follows by iterated application of Theorem 4.3 if all rows with index greater than p are empty. Suppose that this is not the case. Let $v = w_{\mathtt{trim}^p(R)}$. We have that $\ell(w(\uparrow^p v)^{-1}) = \ell(w) - \ell(v)$ by the definition of an RC graph. We may therefore recover the image of R by removing all elements in rows with index greater than p to obtain an RC graph R' while keeping track of $\mathtt{trim}^p(R)$. This reduces the problem to the previous case, where the third coordinate of the tuple becomes $w(\uparrow^p v)^{-1}$ and the first is $\mathtt{clip}^p(R')$, allowing us to uniquely determine R'. Combining this with knowledge of $\mathtt{trim}^p(R)$, we may recover the original graph, as desired. □

Theorem 4.5. *The product $\diamond$ is associative and $\mathcal{BRC}$ is a ring when equipped with $\diamond$.*

Proof. Suppose we have three bounded RC graphs R_1 with $\mathtt{ht}(R_1) = p$, R_2 with $\mathtt{ht}(R_2) = q$, and R_3 with $\mathtt{ht}(R_3) = r$, and we want to show that the order of bracketing the product is irrelevant. Let R be a bounded RC graph with $\mathtt{ht}(R) = p + q + r$. By Lemma 4.4.3, we have that

$$A_{p+q}(R) = (P_1, R_3, w_R)$$

is a unique representation of R, where $\mathtt{ht}(P_1) = p + q$ and $\mathtt{ht}(R_3) = r$. Applying A_p to the first coordinate, we obtain

$$(A_p(P_1), R_3, w_R) = ((R_1, R_2, w_{P_1}), R_3, w_R)$$

where $\mathtt{ht}(R_1) = p$, $\mathtt{ht}(R_2) = q$. We also have

$$A_p(R) = (R_1', P_2, w_R)$$

where P_2 is a unique RC graph with $\mathtt{ht}(P_2) = q + r$. Applying A_q to the second coordinate, we have

$$(R_1', A_q(P_2), w_R) = (R_1', (R_2', R_3', w_{P_2}), w_R)$$

We may immediately conclude that $R_3 = R_3'$. Without loss of generality, we may therefore assume that R_3 is empty. The two representations therefore become

$$((R_1, R_2, w_{P_1}), \emptyset, w_R)$$

and

$$(R_1', (R_2', \emptyset, w_{P_2}), w_R)$$

Consider the change in representation when we clip R to $p+q$ rows. We can see that the first representation becomes simply

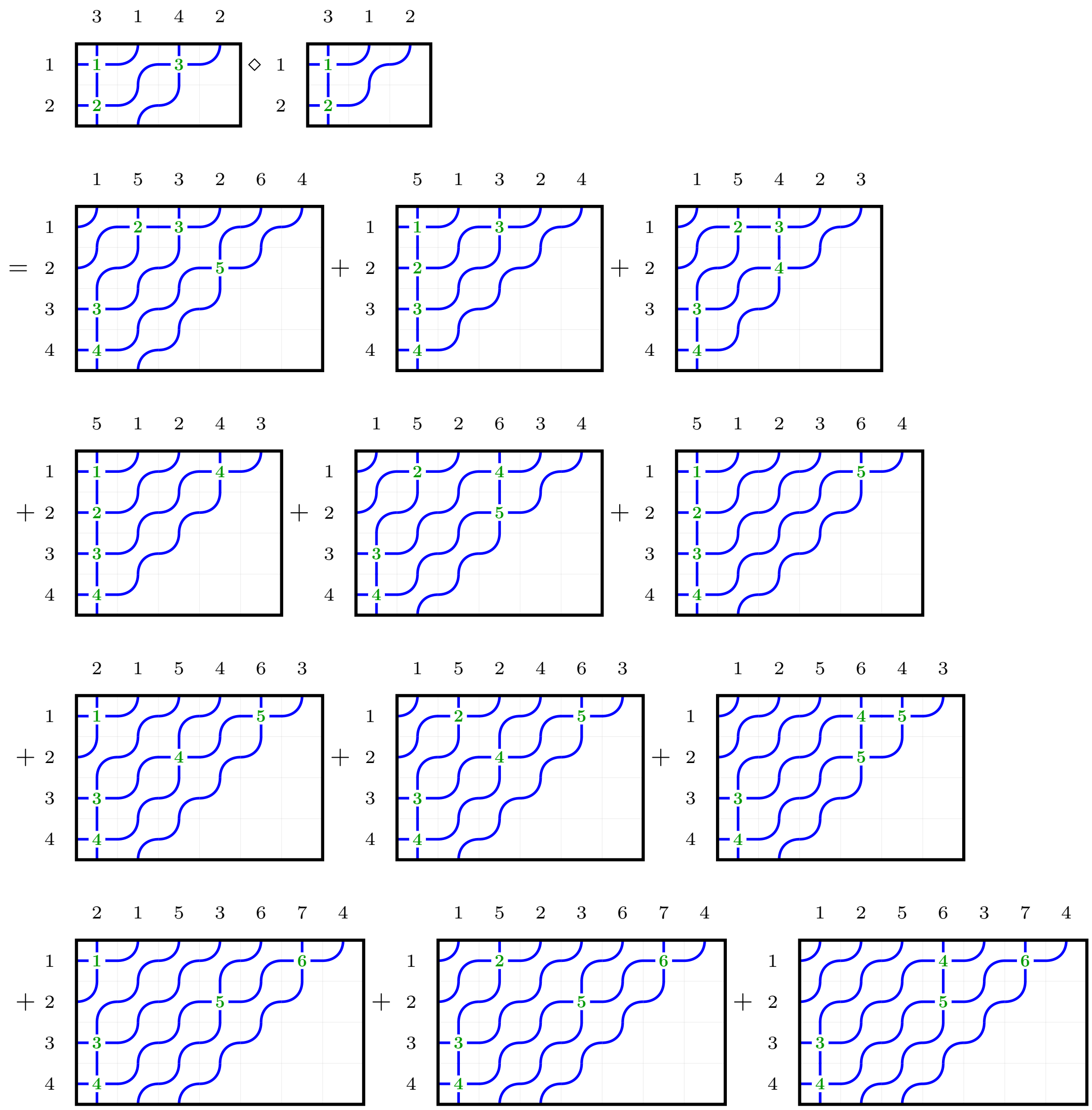
$$(R_1, R_2, w_{P_1})$$

and the RC graph P_2 is clipped in the second representation, obtaining

$$(R_1', R_2', w_R)$$

since $R_2' = \mathtt{clip}^q(P_2)$. These represent the same RC graph $\mathtt{clip}^{p+q}(R)$. It follows that $R_1 = R_1'$ and $R_2 = R_2'$ from Lemma 4.4.3. We may conclude that if we know the RC graphs R_1, R_2, R_3, then the RC graphs that appear as terms in the product $(R_1 \diamond R_2) \diamond R_3 = R_1 \diamond (R_2 \diamond R_3)$ do not depend on the bracketing. □

Example 4.4.4. We compute the product of two bounded RC graphs $(R_1, 2) \in \mathcal{BRC}(2413, 2)$ and $(R_2, 2) \in \mathcal{BRC}(231, 2)$:



4.5. **Proof of the Littlewood-Richardson rule for the dual Schubert basis.**

Definition 4.5.1. Define a map $\Xi : \mathcal{BRC} \to \mathcal{D}$ by

$$\Xi(R) = \Xi_{w_R}^{\mathtt{ht}(R)}$$

and $\omega : \mathcal{BRC} \to \mathcal{D}$ by

$$\omega(R) = [\mathtt{wt}(R)]$$

Let $\mathcal{R}$ be the subring of $\mathcal{BRC}$ generated by the single-row RC graphs $\mathtt{row}(i)$ for $i \geq 0$. Let a be a sequence of elements of a ring indexed by the nonnegative integers. For a permutation w and an integer $n \geq \mathfrak{m}(w)$, define a function $\Xi_w^n(a)$ by

$$\Xi_w^n(a) = \sum_{\alpha} d_{\alpha,w}^n a_{\alpha_1} \cdots a_{\alpha_n}$$

where the sum is over all weak compositions α of length n, and $d_{\alpha,w}^n$ is the coefficient of α in $\Xi_w^n \in \mathcal{D}$.

Proposition 4.5.2. *We have that the elements*

$$\Xi_w^n(\mathtt{row})$$

for integers $n \geq 0$ and w that ranges over all permutations in S_∞ with $n \geq \mathfrak{m}(w)$ form a $\mathbb{Z}$-basis of $\mathcal{R}$, and $\alpha : \mathcal{R} \to \mathcal{D}$ is an isomorphism of graded rings.

Proof. Since $\alpha(\Xi_w(\mathtt{row})) = \Xi_w^n \in \mathcal{D}$, the elements $\Xi_w^n(\mathtt{row})$ are linearly independent. Given that the images form a basis of $\mathcal{D}$, they must also span $\mathcal{R}$ since $\mathcal{D}$ is a free algebra, and hence there is a homomorphism backwards $\mathcal{D} \to \mathcal{R}$ sending $[i] \mapsto \mathtt{row}(i)$ which is necessarily surjective. Thus, the elements $\Xi_w^n(\mathtt{row})$ form a $\mathbb{Z}$-basis of $\mathcal{R}$. $\square$

Definition 4.5.3. Let Ω be the $\mathbb{Z}$-submodule of $\mathcal{BRC}$ generated by the RC graphs $R_1 - R_2$ such that $w_{R_1} = w_{R_2}$ and $\mathtt{ht}(R_1) = \mathtt{ht}(R_2)$.

Lemma 4.5.4. *Ω is a two-sided ideal of $\mathcal{BRC}$.*

Proof. The fact that Ω is a left ideal follows from the fact that a product $R \diamond R_1$ is uniquely determines by $R \diamond \mathrm{empty}(\mathtt{ht}(R_1))$ and R_1, and the contribution of R_1 is precisely that it replaces the last $\mathtt{ht}(R_1)$ rows of each term. In particular, precisely the same $\mathtt{ht}(R)$ occur in precisely the same number of terms, to which $\uparrow^{\mathtt{ht}(R)} R_1$ is added as the last $\mathtt{ht}(R_1)$ rows. It follows that $R \diamond R_1 - R \diamond R_2$ is a linear combination of terms of the form $R_1' - R_2'$ such that $\mathtt{ht}(R_1') = \mathtt{ht}(R_2')$ and $w_{R_1'} = w_{R_2'}$, and Ω is a left ideal.

The fact that Ω is a right ideal follows from the fact that a product $R_1 \diamond R$ is uniquely determined by $R_1 \diamond \mathrm{empty}(\mathtt{ht}(R))$ and R, and the contribution of R is precisely that it replaces the last $\mathtt{ht}(R)$ rows of each term. In particular, precisely the same $\mathtt{ht}(R)$ occur in precisely the same number of terms, to which $\uparrow^{\mathtt{ht}(R)} R$ is added as the last $\mathtt{ht}(R)$ rows. It follows that $R_1 \diamond R - R_2 \diamond R$ is a linear combination of terms of the form $R_1' - R_2'$ such that $\mathtt{ht}(R_1') = \mathtt{ht}(R_2')$ and $w_{R_1'} = w_{R_2'}$, and Ω is a right ideal. $\square$

Theorem 4.6. *We have that $\alpha : \mathcal{BRC} \to \mathcal{D}$ is a surjective homomorphism of rings with kernel Ω, and the homomorphism $\iota : \mathcal{D} \to \mathcal{BRC}$ sending $[i] \mapsto \mathtt{row}(i)$ is a right inverse to α. In particular, $\mathcal{BRC} \cong \mathcal{D} \ltimes \Omega$ as rings.*

Proof. We have an exact sequence

$$0 \to \Omega \to \mathcal{BRC} \to \mathcal{D} \to 0$$

where the map $\mathcal{BRC} \to \mathcal{D}$ is α. We have a homomorphism $\iota : \mathcal{D} \to \mathcal{BRC}$ sending $[i] \mapsto \mathtt{row}(i)$ inducing an isomorphism on $\mathcal{R}$, and the composition $\mathcal{D} \to \mathcal{BRC} \to \mathcal{D}$ is the identity since $\alpha(\mathtt{row}(i)) = [i]$. Since Ω is a two-sided ideal, we can conclude that α is in fact a homomorphism of rings. Since $\mathcal{BRC} = \mathcal{R} \oplus \Omega$, by virtue of the fact that the sequence is split, and $\mathcal{R} \cong \mathcal{D}$, we have $\mathcal{BRC} \cong \mathcal{D} \ltimes \Omega$ as rings. $\square$

Proposition 4.5.5. *Let $[c]$ be a weak composition of length n. Then*

$$[c] = \sum_{\substack{w \in S_\infty \\ \mathfrak{m}(w) \leq n}} \# R \in \mathcal{RC}_n(w) \mid wtt(R) = c \Xi_{w_R}^n$$

Definition 4.5.6. Given RC graphs $U \in \mathcal{RC}_p(u)$ and $V \in \mathcal{RC}_q(v)$ and a permutation w, define

$$d_{U,V}^w = \#\{R \in \mathcal{RC}(w, p+q) : \mathtt{clip}^p(R) = U, \mathtt{trim}^p(R) = V\}$$

Proof of Theorem 1.1. For any choice of RC graphs $U_1, U_2 \in \mathcal{RC}_p(u)$ and $V_1, V_2 \in \mathcal{RC}_q(v)$ and a permutation w, we have $\alpha(U_1) = \alpha(U_2)$ and $\alpha(V_1) = \alpha(V_2)$, so that $\alpha(U_1)\alpha(V_1) = \alpha(U_2)\alpha(V_2)$. We have

$$U_1 \diamond V_1 = \sum_{\mathtt{clip}^p(R) = U_1, \mathtt{trim}^p(R) = V_1} R$$

Applying α, we then have that

$$\Xi_u^p \Xi_v^q = \alpha(U_1)\alpha(V_1) = \alpha(U_1 \diamond V_1) = \sum_{\mathtt{clip}^p(R) = U_1, \mathtt{trim}^p(R) = V_1} \alpha(R) = \sum_{\mathtt{clip}^p(R) = U_1, \mathtt{trim}^p(R) = V_1} \Xi_{w_R}^{p+q}$$

The coefficient of Ξ_w^{p+q} in this product is precisely $d_{U_1,V_1}^w = d_{U_2,V_2}^w$ by definition, and we have the result. $\square$

5. The ring of RC graph crystals

5.1. Little bumps and the crystal action on RC graphs.

Definition 5.1.1 (Crystal action on RC graphs, highest weights, extremal weights)**.** Following Morse and Schilling [22], who defined an $A_{\ell-1}$-crystal structure on reduced factorizations of a permutation w, and the adaptation to RC graphs by Assaf and Schilling [1], we equip $\mathcal{RC}(w)$ with Kashiwara raising and lowering operators e_i, f_i for $i \geq 1$. The operator pair (e_i, f_i) acts only on rows i and $i+1$ of an RC graph R, viewed as adjacent factors in the corresponding reduced factorization.

The action is governed by a bracketing procedure on the column indices appearing in rows i and $i+1$. Process the entries of row i in decreasing order of column index: for each column b in row i, pair it with the smallest column $a > b$ appearing in row $i+1$ that has not yet been paired in an earlier step; if no such a exists, b is left unpaired. Let $R_i(R)$ denote the set of unpaired column indices in row i, and $L_i(R)$ the set of unpaired column indices in row $i+1$.

If $R_i(R)$ is nonempty, $f_i(R)$ is obtained by removing the smallest unpaired entry $b \in R_i(R)$ from row i and inserting an entry into row $i+1$ at the largest column $b' < b$ such that b' does not already appear in row $i+1$; if $R_i(R) = \emptyset$ or no such b' exists, $f_i(R) = \emptyset$. Dually, if $L_i(R)$ is nonempty, $e_i(R)$ is obtained by removing the largest unpaired entry $a \in L_i(R)$ from row $i+1$ and inserting an entry into row i at the smallest column $a' > a$ such that a' does not already appear in row i; if $L_i(R) = \emptyset$ or no such a' exists, $e_i(R) = \emptyset$.

The *weight* of R as an element of the crystal is simply $\mathtt{wt}(R)$, the row weight itself. A *highest weight* element is an RC graph R with $e_i(R) = \emptyset$ for all $i \geq 1$. Every $R \in \mathcal{RC}(w)$ corresponds to a unique highest weight element $\mathbb{Y}(R)$, which can always be obtained by greedily applying available e_i until none apply.The *Demazure crystal* $\operatorname{Dem}(R)$ generated by a highest weight $\mathbb{Y}(R)$ is the closure of $\{\mathbb{Y}(R)\}$ under the operators $\{f_i^k : i \geq 1, k \geq 0\}$ subject to the Demazure truncation rule of [1]. The *extremal weight* $\operatorname{extwt}(R)$ of R is the weak composition that is the lexicographically smallest weight of any element of $\operatorname{Dem}(R)$.

The significance of this elaborate structure is the following.

Theorem 5.1. *If $A \in \mathcal{BRC}$ is such that $\operatorname{extwt}(A) = a$, then*

$$\kappa_a = \sum_{R \in \operatorname{Dem}(A)} x^{\mathtt{wt}(R)}$$

It will be advantageous to have an alternative, crystal-friendly definition of the map $\mathcal{Z}$ for the purposes of proving results about key polynomials.

Definition 5.1.2 (Little bumps)**.** Given a reduced word $a_1 \cdots a_m$ for a permutation w and an index $1 \leq p \leq m$, the *Little bump* $\operatorname{little}_p(a_1 \cdots a_m)$ is defined recursively as follows. If $a_p > 1$, replace a_p with $a_p - 1$, and if $a_p = 1$ then instead replace all a_q with $q \neq p$ with $a_q + 1$. Denote the new word by $a_1' \cdots a_m'$. If this resulting word is reduced, we are done. Otherwise, there is a unique index $j \neq p$ such that the word obtained by deleting a_j' from the new word is reduced by the deletion property of Coxeter groups. We then define

$$\operatorname{little}_p(a_1 \cdots a_m) = \operatorname{little}_j(a_1' \cdots a_m')$$

Definition 5.1.3. Define a function $\mathscr{L} : \mathcal{BRC}^0(n) \to \mathcal{BRC}(n+1)$ as follows. Assume $n = \mathfrak{m}(w_R)$, and suppose $\mathtt{rt}_R(i,j) = (n, n+1)$, where $i < n$. Set

$$R' = (R \setminus \{(i,j)\}) \cup \{(n,1)\}$$

Then define

$$\mathscr{L}(R) = (i \overset{n-1}{\rightsquigarrow} R') \setminus \{(n,1)\}$$

which we assign height $n+1$. Then define $\mathscr{Z} : \mathcal{BRC}^0(n) \to \mathcal{BRC}(n-1)$ by iterating $\mathscr{L}$ and stopping as soon as the result has $\mathfrak{m}(w_R) \leq n-1$, then reducing the number of rows to $n-1$.

Proposition 5.1.4. *Let $(R,n) \in \mathcal{BRC}^0(n)$ be a bounded RC graph such that $\mathfrak{m}(w_R) = n$. Suppose*

$$\operatorname{word}(R) = a_1 a_2 \cdots a_k$$

and

$$\operatorname{seq}(R) = r_1 r_2 \cdots r_k$$

suppose $\mathtt{rt}_R(i,j) = (n, n+1)$*, and let* p *be the index such that* $r_p = i$ *and* $r_p + j - 1 = a_p$*. Then*

$$\mathrm{word}(\mathscr{L}(R)) = \mathrm{little}_p(\mathrm{word}(R))$$

Proof. Consider the biword

$$\begin{matrix} r_1 & r_2 & \cdots & r_{p-1} & r_p & r_{p+1} & \cdots & r_k \\ a_1 & a_2 & \cdots & a_{p-1} & a_p & a_{p+1} & \cdots & a_k \end{matrix}$$

The first deletion yields

$$\begin{matrix} r_1 & r_2 & \cdots & r_{p-1} & r_{p+1} & \cdots & r_k & n \\ a_1 & a_2 & \cdots & a_{p-1} & a_{p+1} & \cdots & a_k & n \end{matrix}$$

Necessarily, we must have $a_p > 1$. To see this, note that if we have $a_p = 1$, then we would have $r_p = 1$ by the compatible sequence condition. The only way for position $(1,1)$ in an RC graph to be the final right descent is if it is the unique right descent. By assumption, that descent would have to be n, so that $n = 1$. In that case, however, it would not be possible for the last row to be empty, which is included in the hypotheses of the proposition.

We must also have that $r_p < a_p$, because there is a sequence of chute moves involving this crossing that moves it to position $(n,1)$ by [3, Theorem 3.7]. Assume first that $p = k$. Then the insertion algorithm will insert $a_p - 1 = n - 1$ into row r_k. If this word is reduced, then we are done, and this clearly coincides with the Little bump. If it is not reduced, the descent $n-1$ occurs somewhere to the left, say at position j. The insertion algorithm would delete this letter then try to insert another one. Recursively, this is the same as applying $\mathscr{L}$ to the biword obtained by removing all letters after position j and then reinserting them afterwards. By the inductive hypothesis, this coincides with applying the Little bump at position j, which would indeed be the next step in the recursion of applying the Little bump at position k. Thus, the two coincide by induction. □

We invoke the following theorem of Hamaker and Young.

Theorem 5.2 ([11, Theorem 4.4])**.** *Let* w *be a reduced word for a permutation* $w \in S_\infty$*. Then*

$$Q(\mathrm{w}) = Q(\mathrm{little}_p(\mathrm{w}))$$

for any valid p*.*

Proposition 5.1.5. *We have that*

$$\mathscr{X} : \mathcal{RC}(w,n)^0 \to \bigcup_{w \overset{n}{\searrow} w'} \mathcal{RC}(w', n-1)$$

is an isomorphism of crystal graphs.

Proof. The fact that this is a weight-preserving bijection follows directly from repeated application of [18, Lemma 7], which states that the Little bump on the last descent realizes the Lascoux-Schützenberger transition formula. The crystal isomorphism follows from Theorem 5.2. □

Theorem 5.3. *We have the equality* $\mathscr{X} = \mathcal{Z}$*.*

Proof. Let R be a bounded RC graph, say with permutation w, with r rows and last descent r such that row r is empty. We prove this by induction on $\ell(w)$ as well as on $s - r$, where $s > r$ is the maximal index such that $w(r) > w(s)$. If $s = r+1$, then the definitions of the two maps are identical.

By the exchange property of Coxeter groups, there is a unique pair (i,j) such that $\mathtt{rt}_R(i,j) = (r, r+1)$. Set

$$R' = R \setminus \{(i,j)\}$$

and assume that $\mathtt{ht}(R') = r+1$. By the inductive hypothesis, we have $\mathscr{X}(R') = \mathcal{Z}(R')$. We have thus established that $\mathscr{X}$ and $\mathcal{Z}$ induce the same bijection between $\mathcal{RC}(ws_r)^0$ and

$$\bigcup_{ws_r \overset{r+1}{\searrow} w'} \mathcal{RC}(w')$$

Inspection of both algorithms reveals that the removal of this crossing and the subsequent reinsertion in the first step of both algorithms is precisely the same, and holding this fixed then applying the inductive hypothesis yields the result. □

Corollary 5.1.6. *For any bounded RC graph R and any $i \geq 1$, we have*

$$\mathcal{Z}(e_i(R)) = e_i(\mathcal{Z}(R))$$

and

$$\mathcal{Z}(f_i(R)) = f_i(\mathcal{Z}(R))$$

whenever the left-hand sides are defined.

5.2. **The dual key basis and the ring of RC graph crystals.**

Definition 5.2.1 (The dual key basis)**.** For a weak composition a of length n, define $\kappa_a^* \in \mathcal{D}_n$ to be the unique element such that $\langle \kappa_a^*, \kappa_b \rangle = \delta_{a,b}$ for all weak compositions b.

Theorem 5.4. *Let c be a weak composition. Then dual key polynomial κ_c^* is equal to*

$$\kappa_c^* = \sum_{\substack{R=\mathbb{Y}(R)\\ \text{extwt}(R)=c}} \Xi_{w_R}^{\mathtt{ht}(R)}$$

Proof. The statement of the theorem is equivalent to Lascoux and Schützenberger's formula for Schubert polynomials in terms of key polynomials through Theorem 5.1. □

Proposition 5.2.2. *Let c be a weak composition of length p. Let $\mathcal{RC}_{\mathbb{Y}}(c)$ be the set of RC graphs R with $\mathtt{wt}(R) = c$ such that* $\text{word}(\mathbb{Y}(R))$ *is minimal in lexicographical order among all RC graphs R' such that* $\text{extwt}(R) = \text{extwt}(R')$. *Then*

$$[c] = \sum_{R\in\mathcal{RC}_{\mathbb{Y}}(c)} \kappa^*_{\text{extwt}(\mathbb{Y}(R))}$$

Definition 5.2.3 (The submodule $\Theta \subseteq \mathcal{BRC}$)**.** Let Θ be the $\mathbb{Z}$-submodule of $\mathcal{BRC}$ generated by the elements $R_1 - R_2$ such that $\mathbb{Y}(R_1) = \mathbb{Y}(R_2)$.

Lemma 5.2.4. *Θ is a two-sided ideal of $\mathcal{BRC}$.*

Proof. The fact that Θ is a left ideal follows from the fact that the products $R \diamond R_1$ and $R \diamond R_2$ have the same first $\mathtt{ht}(R)$ rows, and the rows below are simply shifts of R_1 and R_2. Since the crystal raising/lowering operators act directly on adjacent rows, the same crystal operators will apply to the rows below, so that $R\diamond R_1$ and $R\diamond R_2$ have the same crystal structure in the trailing rows. In passing to highest weights $\mathbb{Y}(R\diamond R_1)$ and $\mathbb{Y}(R\diamond R_2)$, we may pass through the partial highest weights $R\diamond Y(R_1) = R\diamond Y(R_2)$, so that in particular the highest weights of the bijectively corresponding terms of $R \diamond R_1$ and $R \diamond R_2$ are the same.

To prove that Θ is a right ideal, the same argument applies, but we must invoke Corollary 5.1.6 to guarantee that the crystals remain isomorphic as needed. □

Lemma 5.2.5. *The homomorphism $\alpha : \mathcal{BRC} \to \mathcal{D}$ factors through the quotient $\mathcal{BRC}/\Theta$.*

Proof. This is because the generators of Θ are all in Ω. □

Definition 5.2.6. For a composition a, define an element $K_a \in \mathcal{BRC}/\Theta$ by

$$K_a = \sum_{\substack{R\in\text{Dem}_{\mathcal{RC}}(a)\\ R=\mathbb{Y}(R)}} R$$

Proposition 5.2.7. *The elements $K_a \in \mathcal{BRC}/\Theta$ form an additive basis of a subring isomorphic to $\mathcal{D}$ under the homomorphism $\alpha : \mathcal{BRC}/\Theta \to \mathcal{D}$, and*

$$\alpha(K_a) = \kappa_a^*$$

Proof. The last statement is simply Theorem 5.4. The K_a actually consist of pairwise disjoint sets of RC graphs, the set of all such is clearly independent. To show that they span a subring, consider the homomorphism $\alpha : \mathcal{BRC}/\Theta \to \mathcal{D}$. We have that $\alpha(K_a) = \kappa_a^*$, and since these are linearly independent, it follows that the kernel of α intersects trivially with the span of the K_a. Since α is a homomorphism, it follows that α is injective on the span of the K_a, and since

$$\alpha(K_a)\alpha(K_b) = \kappa_a^*\kappa_b^* = \sum_c k_{ab}^c \kappa_c^* = \sum k_{a,b}^c \alpha(K_c),$$

it follows that K_aK_b is a linear combination of the K_c, so the span of the K_a is closed under multiplication, so that α is an isomorphism and we are done. □

Proof of Theorem 1.2. The coefficient of K_c in K_aK_b is the same as the coefficient of κ_c^* in $\kappa_a^*\kappa_b^*$, which is $k_{a,b}^c$ by definition. Since each of the summands of K_c that are Demazure crystals of RC graphs are distinct from the summands of any other key element and each other, we need only consider one of the summands to read off the coefficient. The result follows. ☐

6. The ring of forest classes of RC graphs

6.1. Indexed forests, forest invariant.

Definition 6.1.1 (Indexed forests and codes)**.** An *indexed forest* F is a pair $(S, T_\bullet)$ where S is a set of positive integers, called the *support* of F, and if we write

$$S = I_1 \cup I_2 \cup \cdots \cup I_k$$

where $I_1, I_2, \ldots, I_k$ are the maximal intervals contained in S, then $T_\bullet$ is a sequence of binary trees $T_1, T_2, \ldots, T_k$ such that T_j has $|I_j|$ internal nodes for each j. The internal nodes of F are denoted by $\mathrm{IN}(F)$. For $v \in \mathrm{IN}(F)$, we denote by $\mathrm{left}(v)$ and $\mathrm{right}(v)$ the left and right children of v respectively, which may not be in $\mathrm{IN}(F)$ if they are leaves, or may be empty.

There is a natural labeling $\lambda_F : \mathrm{IN}(F) \to S$ sending each internal node, say $v \in T_j$, to the unique element of I_j that is at the same index as $v \in T_j$ in the inorder traversal of T_j.

The function $\rho_F : \mathrm{IN}(F) \to S$ is defined for an internal node $v \in T_j$ by

$$\rho_F(v) = \begin{cases} \lambda_F(v) & \text{if } \mathrm{left}(v) \notin \mathrm{IN}(F) \\ \lambda_F(\mathrm{left}(v)) & \text{if } \mathrm{left}(v) \in \mathrm{IN}(F). \end{cases}$$

For each indexed forest F, we assign a weak composition $\mathfrak{c}(F)$ by

$$\mathfrak{c}_j(F) = \#\{v \in \mathrm{IN}(F) : \rho_F(v) = j\}$$

Then the *forest polynomial* $\mathfrak{P}_F(x)$ is defined by

$$\mathfrak{P}_F(x) = \sum_{\tau} \prod_{v \in \mathrm{IN}(F)} x_{\tau(v)}$$

where the sum is over all functions $\tau : \mathrm{IN}(F) \to \mathbb{Z}_{>0}$ such that $\tau(v) \leq \rho_F(v)$ for all $v \in \mathrm{IN}(F)$, $\tau(v) \leq \tau(\mathrm{left}(v))$ for all $v \in \mathrm{IN}(F)$ such that $\mathrm{left}(v) \in \mathrm{IN}(F)$, and $\tau(v) < \tau(\mathrm{right}(v))$ for all $v \in \mathrm{IN}(F)$ such that $\mathrm{right}(v) \in \mathrm{IN}(F)$.

It is easiest for our purposes to index forest polynomials by weak compositions rather than indexed forests, so we define $\mathfrak{P}_a(x) = \mathfrak{P}_F(x)$ where F is the unique indexed forest such that $\mathfrak{c}(F) = a$.

Definition 6.1.2 (Injective words, forest invariant)**.** In [24], an alphabet $\overline{\mathbb{Z}}$ is defined as the set of pairs of integers (i, j) in lexicographical order, written with the notation $i^{[j]}$. They define $\mathrm{val}(i^{[j]}) = i$. An *injective word* is a word $w = w_1w_2 \cdots w_k$ such that $w_i \neq w_j$ for all $i \neq j$. An *LBS (local binary search) labeling* of an indexed forest F is an injective function $\rho : \mathrm{IN}(F) \to \overline{\mathbb{Z}}$ such that

(1) $\rho(\mathrm{left}(v)) < \rho(v)$ for all $v \in \mathrm{IN}(F)$ such that $\mathrm{left}(v) \in \mathrm{IN}(F)$.
(2) $\rho(v) < \rho(\mathrm{right}(v))$ for all $v \in \mathrm{IN}(F)$ such that $\mathrm{right}(v) \in \mathrm{IN}(F)$.
(3) For all $v \in \mathrm{IN}(F)$, $\mathrm{val}(\rho(v)) = \lambda_F(u)$ for some $u \in \mathrm{IN}(F)$.

Following [23], given any injective word $w = w_1 \cdots w_m$ in $\overline{\mathbb{Z}}$, we associate to it inductively an indexed forest $F(w)$, an LBS labeling $P(w)$ of $F(w)$, and an injective "time-stamp" labeling $Q(w) : \mathrm{IN}(F(w)) \to \{1, \ldots, m\}$ as follows. If $m = 0$, then $F(w)$ is the empty forest. Otherwise write $w = w'a$ with $w' = w_1 \cdots w_{m-1}$, and assume $F' := F(w')$, $P' := P(w')$, and $Q' := Q(w')$ have already been constructed; let S' be the support of F', and let $S' = I_1 \cup \cdots \cup I_k$ be its decomposition into maximal intervals (listed left to right). For each j, denote by u_j the root of the tree of F' supported on I_j, and set $a_j := P'(u_j)$. We obtain $F(w)$ by adjoining a new internal node u to F', with attachments determined by $i := \mathrm{val}(a)$:

- If $i \notin S'$, then u is the root of a new tree, with left child u_j when $i - 1 \in I_j$ (otherwise no left child), and right child u_j when $i + 1 \in I_j$ (otherwise no right child).
- If $i \in I_j$, then we compare a with a_j in the order on $\overline{\mathbb{Z}}$:
  - if $a > a_j$, then u_j becomes the left child of u; additionally, if $\max(I_j) + 2 \in S'$ (necessarily $\max(I_j) + 2 = \min(I_{j+1})$), then u_{j+1} becomes the right child of u;

– if $a < a_j$, then u_j becomes the right child of u; additionally, if $\min(I_j) - 2 \in S'$ (necessarily $\min(I_j) - 2 = \max(I_{j-1})$), then u_{j-1} becomes the left child of u.

The remaining roots of F' become roots of $F(w)$ unchanged. We then extend P' to $P(w)$ by setting $P(w)(u) := a$, and extend Q' to $Q(w)$ by setting $Q(w)(u) := m$. The pair $(P(w), Q(w))$ is the Ω-*insertion* of w.

The importance of this algorithm is

Theorem 6.1 ([23, Theorem 5.1])**.** *The correspondence $w \mapsto (P(w), Q(w))$ is a bijection between the following two sets:*

(1) *Injective words w with letters in $\overline{\mathbb{Z}}$, and*
(2) *Pairs (P, Q) such that $P \in \mathrm{LBS}(F)$ and $Q \in \mathrm{Dec}(F)$ for a common indexed forest F.*

Definition 6.1.3 (Injectification, $\equiv_{\mathfrak{P}}$)**.** For a word $\mathbf{r}$ in the alphabet $\mathbb{Z}$, we may canonically turn $\mathbf{r}$ into an injective word in the alphabet $\overline{\mathbb{Z}}$ by assigning superscripts consecutively starting at 1 to break ties between equal letters.

Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be reduced words. Then we say that $\mathbf{r}_1 \equiv_{\mathfrak{P}} \mathbf{r}_2$ if $P(\overline{\mathbf{r}_1}) = P(\overline{\mathbf{r}_2})$. Naturally, we may apply this to RC graphs as well: for RC graphs R_1 and R_2, we say that $R_1 \equiv_{\mathfrak{P}} R_2$ if $\overline{\mathrm{word}(R_1)} \equiv_{\mathfrak{P}} \overline{\mathrm{word}(R_2)}$. For a permutation w, we denote by $\mathcal{C}_{\mathfrak{P}}(w)$ the set of equivalence classes of reduced words for w under $\equiv_{\mathfrak{P}}$.

Theorem 6.2 ([23, Theorem 1.4])**.** *Let $w \in S_\infty$. Then we have*

$$\mathfrak{S}_w(x) = \sum_{C \in \mathcal{C}_{\mathfrak{P}}(w)} \mathfrak{P}_{\mathfrak{c}_{\mathfrak{P}}(C)}(x)$$

6.2. **The dual forest basis and the ring of forest classes of RC graphs.**

Definition 6.2.1 (The dual forest basis)**.** For a weak composition a of length n, define $\mathfrak{P}^*_a \in \mathcal{D}_n$ to be the unique element such that $\langle \mathfrak{P}^*_a, \mathfrak{P}_b \rangle = \delta_{a,b}$ for all weak compositions b.

Theorem 6.3. *We have the following.*

I *Let $R \in \mathcal{BRC}$ be a bounded RC graph. Then the forest polynomial $\mathfrak{P}_{\mathfrak{c}_f(R)}$ is equal to*

$$\mathfrak{P}_{\mathfrak{c}_f(R)}(x) = \sum_{R' \equiv_{\mathfrak{P}} R} x^{\mathtt{wt}(R')}$$

II *Let c be a weak composition. Then dual forest polynomial $\mathfrak{P}^*_c$ is equal to*

$$\mathfrak{P}^*_c = \sum_{[R] \in \mathcal{C}_{\mathfrak{P}}(c)} \Xi^{\mathtt{ht}(R)}_{w_R}$$

Proof of Theorem 6.3. Part I of this theorem follows from the fact that for any equivalence class of words W that have the same P-LBS, $\mathfrak{P}_{\mathfrak{c}_{\mathfrak{P}}(W)}(x)$ is the sum of the compatible sequences on the words in W, which are simply the RC graphs in the corresponding equivalence class of RC graphs. Part II follows from Theorem 6.2 and the definition of the dual Schubert basis. $\square$

Guo and Woodruff [10] also published an alternative RC graph formula for forest polynomials.

Proposition 6.2.2. *Let c be a weak composition of length p. Let $\mathcal{RC}_{\mathfrak{P}}(c)$ be the set of RC graphs with $\mathtt{wt}(R) = c$ such that $\mathrm{word}(\mathbb{Y}_{\mathfrak{P}}(R))$ is minimal in lexicographical order among all RC graphs R' such that $\mathfrak{c}_{\mathfrak{P}}(R) = \mathfrak{c}_{\mathfrak{P}}(R')$.*

$$[c] = \sum_{R \in \mathcal{RC}_{\mathfrak{P}}(c)} \mathfrak{P}^*_{\mathfrak{c}_{\mathfrak{P}}(R)}$$

Definition 6.2.3 (The submodule $\Gamma \subseteq \mathcal{BRC}$)**.** Let Γ be the $\mathbb{Z}$-submodule of $\mathcal{BRC}$ generated by the elements $R_1 - R_2$ such that $R_1 \equiv_{\mathfrak{P}} R_2$.

Lemma 6.2.4. Γ *is a two-sided ideal of $\mathcal{BRC}$.*

Proof. The fact that Γ is a left ideal follows from the fact that the products $R \diamond R_1$ and $R \diamond R_2$ have the same first $\mathtt{ht}(R)$ rows, and the rows below are simply shifts of R_1 and R_2. In terms of the resulting words, the shifted word(R_1) and word(R_2) are suffixes of word$(R \diamond R_1)$ and word$(R \diamond R_2)$. The $\equiv_{\mathfrak{P}}$ equivalence is determined by local swaps of pairs of letters that are related in a shift-invariant [23, Proposition 5.8] and prefix-invariant [23, Lemma 5.9] way, from which the result follows, since the prefixes are identical and the suffixes are related in the same way as word(R_1) and word(R_2).

To prove that Γ is a right ideal, assume word(R_1) and word(R_2) differ by a single local swap of adjacent letters a and b in the same equivalence class, with a to the left of b in word(R_1) and to the right of b in word(R_2). It suffices to consider only the case where R is empty to show that

$$R_1 \diamond R - R_2 \diamond R \in \Gamma$$

for any R. According to [23], the local swap of a and b is a commutation move, and we may reduce to the case where a and b are the last two letters of word(R_1), and that b is the last descent of w_{R_1}. For any empty R, the position a will remain the same. For any term not identical to R_1, the suffix of the word will be ab' where $b' > b$. For each b', there will be precisely one matching term with suffix $b'a$ and identical prefix. The local swap of a and b' in the suffix is a valid local swap, so the resulting terms are all in Γ, and we are done. ☐

Lemma 6.2.5. *The homomorphism $\alpha : \mathcal{BRC} \to \mathcal{D}$ factors through the quotient $\mathcal{BRC}/\Gamma$.*

Proof. This is because the generators of Γ are all in Ω. ☐

Definition 6.2.6. For a composition a, define an element $F_a \in \mathcal{BRC}/\Gamma$ by

$$F_a = \sum_{[R] \in \mathcal{C}_{\mathfrak{P}}(a)} [R]$$

Proposition 6.2.7. *The elements $F_a \in \mathcal{BRC}/\Gamma$ form an additive basis of a subring isomorphic to $\mathcal{D}$ under the homomorphism $\alpha : \mathcal{BRC}/\Gamma \to \mathcal{D}$, and*

$$\alpha(F_a) = \mathfrak{P}_a^*$$

Proof. As before, the last statement is simply Theorem 6.3. The F_a actually consist of pairwise disjoint sets of RC graphs, the set of all such is clearly independent. To show that they span a subring, consider the homomorphism $\alpha : \mathcal{BRC}/\Gamma \to \mathcal{D}$. We have that $\alpha(F_a) = \mathfrak{P}_a^*$, and since these are linearly independent, it follows that the kernel of α intersects trivially with the span of the F_a. Since α is a homomorphism, it follows that α is injective on the span of the F_a, and since

$$\alpha(F_a)\alpha(F_b) = \mathfrak{P}_a^*\mathfrak{P}_b^* = \sum_c f_{ab}^c \mathfrak{P}_c^* = \sum f_{a,b}^c \alpha(F_c),$$

it follows that $F_a F_b$ is a linear combination of the F_c, so the span of the F_a is closed under multiplication, so that α is an isomorphism and we are done. ☐

Proof of Theorem 1.3. The coefficient of F_c in $F_a F_b$ is the same as the coefficient of $\mathfrak{P}_c^*$ in $\mathfrak{P}_a^*\mathfrak{P}_b^*$, which is $f_{a,b}^c$ by definition. Since each of the summands of F_c that are equivalence classes of RC graphs are distinct from the summands of any other forest element and each other, we need only consider one of the summands to read off the coefficient. The result follows. ☐

7. Product rule for dual slide polynomials

7.1. **Slide polynomials.** RC graph is a triple (w, E, α) where w is a permutation,

$$E : I(w) \to [\ell(w)]$$

is a linear extension, and $\alpha : I(w) \to \mathbb{Z}_{>0}$ is a compatible sequence.

Definition 7.1.1. Slide polynomials $\mathfrak{F}(\mathbf{r})$ for a word $\mathbf{r}$ are defined by summing over all compatible sequences of $\mathbf{r}$. They can be characterized as follows.

A *quasi-Yamanouchi* RC graph is an RC graph R such that if R' is another RC graph with word$(R) =$ word(R') and $\mathtt{ht}(R) = \mathtt{ht}(R')$, then $\mathtt{wt}(R) \leq \mathtt{wt}(R')$ in dominance order and flat$(\mathtt{wt}(R'))$ refines flat$(\mathtt{wt}(R))$.

The set of RC graphs A such that $\mathrm{word}(A) = \mathrm{word}(R)$ and $\mathtt{ht}(A) = \mathtt{ht}(R)$ is a partially ordered set with R as it's smallest element. We then define

$$\mathbb{QY}(A) = R$$

Theorem 7.1 ([2, Theorem 3.13])**.** *For a permutation $w \in S_\infty$, we have the formula*

$$\mathfrak{S}_w(x) = \sum_R \mathfrak{F}_{\mathtt{wt}(R)}(x)$$

where the sum is over all quasi-Yamanouchi RC graphs R for w.

Corollary 7.1.2. *For a weak composition c of length p, we have the formula*

$$\mathfrak{F}_c^* = \sum_{R=\mathbb{QY}(R)} \Xi^p_{w_R}$$

where the sum is over all quasi-Yamanouchi RC graphs R such that $\mathtt{wt}(R) = c$.

$$\sum_{\substack{R\in\mathcal{RC}(c)\\R=\mathbb{QY}(R)}} \Delta(\Xi^p_{w_R}) = \Delta(\mathfrak{F}_c^*)$$

$$\sum_{\substack{R\in\mathcal{RC}(c)\\R=\mathbb{QY}(R)}} c^{w_R}_{u,v}\Xi^p_u \otimes \Xi^p_v = \mathrm{shuff}^c_{a,b}\mathfrak{F}_a^* \otimes \mathfrak{F}_b^*$$

$$\sum_{\substack{R\in\mathcal{RC}(c)\\R=\mathbb{QY}(R)}} c^{w_R}_{u,v}\Xi^p_u \otimes \Xi^p_v = \mathrm{shuff}^c_{a,b} \sum_{\substack{A\in\mathcal{RC}(u,a)\\A=\mathbb{QY}(A)}} \sum_{\substack{B\in\mathcal{RC}(v,b)\\B=\mathbb{QY}(B)}} \Xi^p_u \otimes \Xi^p_v$$

$$\sum_{\substack{R\in\mathcal{RC}(c)\\R=\mathbb{QY}(R)}} c^{w_R}_{u,v} = \mathrm{shuff}^c_{a,b}\#\{A \in \mathcal{RC}(u,a) \mid A = \mathbb{QY}(A)\}\#\{B \in \mathcal{RC}(v,b) \mid B = \mathbb{QY}(B)\}$$

Definition 7.1.3. For a weak composition c, define $\mathcal{RC}_{\mathbb{QY}}(c)$ to be the set of RC graphs with $\mathtt{wt}(R) = c$ such that $\mathrm{word}(\mathbb{QY}(R))$ is minimal in lexicographical order among all RC graphs R' with $\mathtt{wt}(\mathbb{QY}(R')) = \mathtt{wt}(\mathbb{QY}(R))$.

Proposition 7.1.4. *Let c be a weak composition of length p.*

$$[c] = \sum_{R\in\mathcal{RC}_{\mathbb{QY}}(c)} \mathfrak{F}^*_{\mathtt{wt}(\mathbb{QY}(R))}$$

Theorem 7.2. *Let a, b be weak compositions of lengths p and q. Write*

$$\mathfrak{F}_a^*\mathfrak{F}_b^* = \sum_c s^c_{a,b}\mathfrak{F}_c^*$$

For each c, fix an RC graph C with $\mathtt{wt}(\mathbb{QY}(C)) = c$. Then $s^c_{a,b}$ is the number of distinct RC graphs R with $\mathrm{word}(R) = \mathrm{word}(C)$ *such that $\mathtt{wt}(\mathbb{QY}(\mathtt{clip}^p(R))) = a$ and $\mathtt{wt}(\mathbb{QY}(\mathtt{trim}^p(R))) = b$.*

Appendix A. Proof of Proposition 2.1.4

We dedicate this section to the otherwise distracting proof of Proposition 2.1.4.

Proposition A.1 (Special case of Pieri formula [29, Theorem 7.1])**.** *Suppose $k \geq 1$ and $u, w \in S_\infty$. If $u \overset{k}{\not\to} w$, then*

$$\delta^w_u \prod_{i=1}^k (x_1 - y_i) = 0$$

If $u \xrightarrow{k} w$, define

$$Q = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$

then

$$\delta^w_u \prod_{i=1}^k (x_1 - y_i) = \prod_{q\in Q}(x_1 - y_q)$$

Proof. This is simply a change of variables from the original theorem. □

Proof of Proposition 2.1.4. Suppose $v \in S_n$. We have

$$\mathfrak{S}_v(x;y) = \delta^{vw_0(n)}(\mathfrak{S}_{w_0(n)}(x;y))$$

This is equal to

$$\delta^{vw_0(n)}\left(\mathfrak{S}_{s_{n+1-i}\cdots s_1 w_0(n)}(x^{(i)};y)\prod_{j=1}^{n+1-i}(x_i - y_j)\right)$$

and, applying the Leibniz formula, is also equal to

$$\sum_{\substack{v' \in S_\infty \\ \ell(v'w_0(n)s_1\cdots s_{n+1-i}) = \ell(w_0(n)s_1\cdots s_{n+1-i}) - \ell(v')}} \mathfrak{S}_{v'}(x^{(i)};y)\delta^{vw_0(n)}_{v'w_0(n)s_1\cdots s_{n+1-i}}\prod_{j=1}^{n+1-i}(x_i - y_j)$$

By the restricted Pieri formula (Proposition A.1), for this to be nonzero necessarily $v'w_0(n)s_1\cdots s_{n+1-i} \xrightarrow{n+1-i} vw_0(n)$. We note that $v'w_0(n)s_1\cdots s_{n+1-i} \xrightarrow{n+1-i} vw_0(n)$ if and only if

$$v \overset{i}{\Rightarrow} v'w_0(n)s_1\cdots s_{n+1-i}w_0(n) = v's_n s_{n-1}\cdots s_i$$

We have that $v's_n \cdots s_i$ is exactly $\varphi_{i,n}(v')$ since $v' \in S_n$, so we require that

$$v \overset{i}{\Rightarrow} \varphi_{i,n}(v')$$

so the sum is over all v' with $v \overset{i}{\searrow} v'$.

Applying Proposition A.1, we obtain that the result is equal to

$$\sum_{v \overset{i}{\searrow} v'} \mathfrak{S}_{v'}(x^{(i)};y)\prod_{a \in A(v',v)}(x_i - y_a)$$

where $A(v',v)$ is the set of all $vw_0(n)(j)$ such that $1 \leq j \leq n+1-i$ and $v'w_0(n)s_1\cdots s_{n+1-i}(j) = vw_0(n)(j)$. These values are the same as at the indices that comprise the set of all $1 \leq j \leq n+1-i$ such that

$$v'(n+2-s_1\cdots s_{n+1-i}(j)) = v(n+2-j)$$

Applying the $s_1\cdots s_{n+1-i}$ to j, since $1 \leq j \leq n+1-i$ we have that

$$s_1\cdots s_{n+1-i}(j) = j+1$$

Hence we need

$$v'(n+2-(j+1)) = v'(n+1-j) = v(n+2-j)$$

Replacing j with $n+2-p$, the indices are the set of all p such that $i < p \leq n+1$ and

$$v'(p-1) = v(p)$$

Thus $A(v',v)$ is the set of all $v(p)$ such that $p > i$ and $v'(p-1) = v(p)$, which is exactly $Q_i(v',v)$, and we are done. □

References

[1] Assaf, S., and Schilling, A. A Demazure crystal construction for Schubert polynomials. *Algebraic Combinatorics 1*, 2 (2018), 225–247.

[2] Assaf, S., and Searles, D. Schubert polynomials, slide polynomials, Stanley symmetric functions and quasi-Yamanouchi pipe dreams. *Advances in Mathematics 306* (2017), 89–122.

[3] Bergeron, N., and Billey, S. RC-graphs and Schubert polynomials. *Experimental Math. 2*, 4 (1993), 257–269.

[4] Bergeron, N., and Sottile, F. Schubert polynomials, the Bruhat order, and the geometry of flag manifolds. *Duke Math. J. 95*, 2 (1998), 373–423.

[5] Bergeron, N., and Sottile, F. Skew Schubert functions and the Pieri formula for flag manifolds. *Trans. Amer. Math. Soc. 354*, 2 (2001), 651–673.

[6] Billey, S. C., Jockusch, W., and Stanley, R. P. Some Combinatorial Properties of Schubert Polynomials. *J. Algebraic Combin. 2*, 4 (1993), 345–374.

[7] Demazure, M. Désingularisation des variétés de Schubert généralisées. *Annales scientifiques de l'École Normale Supérieure 7*, 1 (1974), 53–88.

[8] Demazure, M. Une nouvelle formule des caractères. *Bulletin des Sciences Mathématiques. 2e Série 98*, 3 (1974), 163–172.
[9] Gao, Y., and Huang, D. The canonical bijection between pipe dreams and bumpless pipe dreams. *International Mathematics Research Notices 2023*, 21 (November 2023), 18629–18663.
[10] Guo, A., and Woodruff, D. Forest polynomials and pattern avoidance, 2026. arXiv:2602.04036 [math.CO].
[11] Hamaker, Z., and Young, B. Relating Edelman–Greene insertion to the Little map. *Journal of Algebraic Combinatorics 40*, 3 (2014), 693–710.
[12] Joseph, A. On the Demazure character formula. *Annales scientifiques de l'École Normale Supérieure 18*, 3 (1985), 389–419.
[13] Kashiwara, M. The crystal base and Littelmann's refined Demazure character formula. *Duke Mathematical Journal 71*, 3 (1993), 839–858.
[14] Knutson, A., and Miller, E. Gröbner geometry of Schubert polynomials. *Annals of Mathematics* (2005), 1245–1318.
[15] Lam, T., Lee, S. J., and Shimozono, M. Back stable Schubert calculus. *Compositio Mathematica 157*, 5 (2021), 883–962.
[16] Lascoux, A., and Schützenberger, M.-P. Keys & standard bases. In *Invariant theory and tableaux (Minneapolis, MN, 1988)*, vol. 19 of *IMA Vol. Math. Appl.* Springer, New York, 1990, pp. 125–144.
[17] Littelmann, P. Crystal graphs and Young tableaux. *Journal of Algebra 175*, 1 (1995), 65–87.
[18] Little, D. P. Combinatorial aspects of the Lascoux–Schützenberger tree. *Advances in Mathematics 174*, 2 (2003), 236–253.
[19] Macdonald, I. G. *Notes on Schubert Polynomials*. Université Du Québec à Montréal, Dép. de mathématiques et d'informatique, 1991.
[20] Mason, S. An explicit construction of type A Demazure atoms. *Journal of Algebraic Combinatorics 29*, 3 (2009), 295–313.
[21] Mathieu, O. Filtrations of G-modules. *Annales scientifiques de l'École Normale Supérieure 23*, 4 (1990), 625–644.
[22] Morse, J., and Schilling, A. Crystal approach to affine Schubert calculus. *International Mathematics Research Notices 2016*, 8 (2016), 2239–2294.
[23] Nadeau, P., and Tewari, V. Forest polynomials and the class of the permutahedral variety. *Advances in Mathematics 453* (2024), 109834.
[24] Nadeau, P., and Tewari, V. P-partitions with flags and back stable quasisymmetric functions. *Combinatorial Theory 4*, 2 (2024). hal-04053201.
[25] Polo, P. Variétés de Schubert et excellentes filtrations. *Astérisque*, 173–174 (1989), 281–311.
[26] Postnikov, A., and Stanley, R. P. Chains in the Bruhat order. *Journal of Algebraic Combinatorics 29*, 2 (2009), 133–174.
[27] Reiner, V., and Shimozono, M. Key polynomials and a flagged Littlewood-Richardson rule. *Journal of Combinatorial Theory, Series A 70*, 1 (1995), 107–143.
[28] Samuel, M. J. *The Leibniz formula for divided difference operators associated to Kac-Moody root systems*. PhD thesis, Rutgers, The State University of New Jersey, 2014.
[29] Samuel, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra 228*, 7 (2024), 107636.
[30] Samuel, M. J. `schubmult` python package. `git@github.com:matthematics/schubmult.git`. Version 4.0.0dev. Stable version available on PyPI., 2026-05-03.
[31] Sottile, F. Pieri's rule for flag manifolds and Schubert polynomials. *Ann. Inst. Fourier 46*, 1 (1996), 89–110.
[32] van der Kallen, W. Longest weight vectors and excellent filtrations. *Mathematische Zeitschrift 201*, 1 (1989), 19–31.
[33] Weigandt, A. Bumpless pipe dreams and alternating sign matrices. *Journal of Combinatorial Theory, Series A 182* (August 2021), 105470.
[34] Yu, T. Embedding Bumpless Pipedreams as Bruhat Chains. *International Mathematics Research Notices 2025*, 22 (2025), rnaf336.